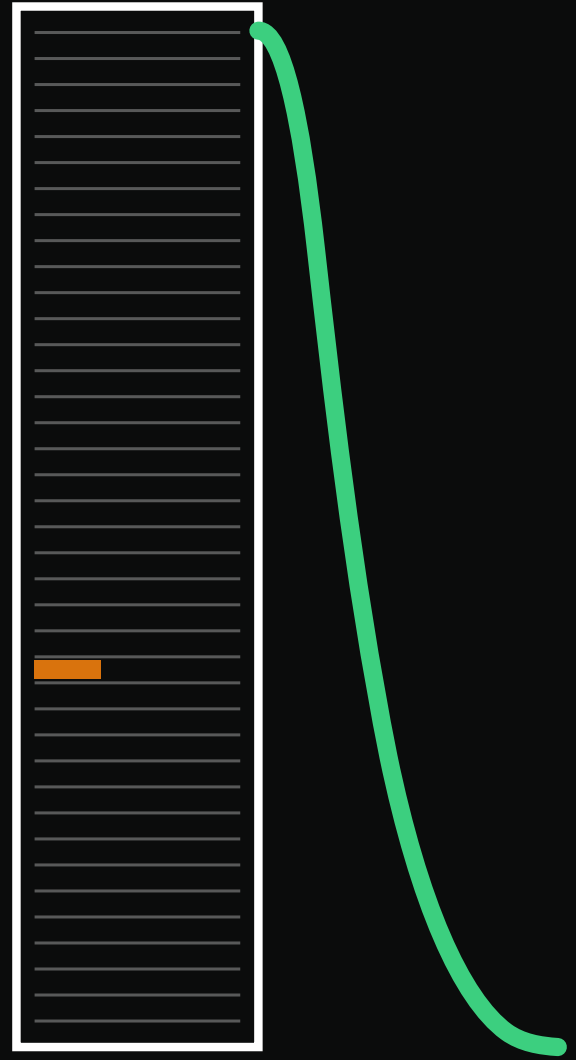


USER GUIDE

The plain-language guide

What TailSafe does, why it matters, and how to use it, step by step and in simple words.



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How to use this guide

This guide explains TailSafe from the very beginning. You do not need to know anything about fire safety, statistics or computers. Every new word is explained the first time it appears, and the [word list](#) at the end explains them all again.

✓ TAILSAFE IN ONE MINUTE

1. People leave tall buildings by the stairs. When a fire starts at night, many people are asleep, some are old or use a wheelchair, and a staircase can fill with smoke.
2. On most nights everyone gets out in good time. On a few bad nights it takes much, much longer. Those few bad nights are the ones that matter.
3. TailSafe plays out hundreds or thousands of possible nights on a computer, each one a little different.
4. It shows how bad the worst nights are, who is still inside, and what causes it.
5. Then it tests cheap changes, such as using lifts for people in wheelchairs or asking neighbours to knock on doors, and tells you honestly what helps and what makes things worse.

How the guide is organised

- **Part 1, The background:** the problem TailSafe was built for, and why the average hides the danger.
- **Part 2, How TailSafe works:** what happens inside, one idea at a time, with pictures and no maths.
- **Part 3, Getting TailSafe running:** the browser version and the full app, click by click.
- **Part 4, Screen by screen:** every one of the nine screens: what it is for, what to press, how to read it.
- **Part 5, Putting it together:** a worked example, the command line, the limits, troubleshooting, questions and the word list.

If you are in a hurry

- To try TailSafe now, on any phone or computer, read [chapter 16](#) and then follow Part 4.
- To understand the one idea behind it all, read [chapter 3](#).
- To run TailSafe for your own building, read [chapter 15](#) and [chapter 17](#).

The boxes

IN PLAIN WORDS

The idea again, in the simplest words possible.

TRY IT

Something to do in TailSafe right now.

WATCH OUT

A common mistake, or a limit to keep in mind.

REMEMBER

The one thing to take away.

Two small conventions

Floors are written the Hong Kong way. G/F is the ground floor, 1/F is the first floor above it, 14/F is the fourteenth floor above the ground. A 40-storey building has the ground floor plus 39 floors above it, so its floors run from G/F to 39/F.

Buttons and labels are printed in bold, exactly as they appear on the screen, for example **Run stress test.**

The background

Before touching any buttons, it helps to understand the problem TailSafe was built for. This part needs no computer at all.

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Tall buildings, fires and people

Many people in Hong Kong live very high up. When there is a fire, almost all of them have to walk down the stairs.

Hong Kong lives up high

Hong Kong has a great many tall homes. Public housing blocks and private towers of 30 to 50 storeys, and more, are ordinary. Well over a thousand people can live in one building, and they share only a few staircases, often just two.

The building this guide uses as its main example is a 40-storey public housing block shaped like a cross. In TailSafe's model it has 760 flats, 2 staircases (called Stair A and Stair B), 3 lifts, and on a Sunday night at 3 a.m. about 1,825 residents at home. The building and its residents are made up by TailSafe; they are not a real block.

How people leave a tall building in a fire

An evacuation (everyone leaving a building because of danger) happens in a few stages:

- 1 **The alarm sounds.**
- 2 **People notice and decide to go.** People who are awake notice quickly. People who are asleep take much longer. The time between the alarm and starting to move is called the **pre-movement time**.
- 3 **They get ready.** They wake the family, find shoes and keys, and help a grandparent out of bed.
- 4 **They walk to a staircase** along the corridor, sometimes through a lobby.
- 5 **They walk down**, floor after floor. On a long staircase people get tired and slow down, and queues form where the staircase is full.
- 6 **They leave through an exit** on the ground floor.

The lifts are normally not used to escape. In Hong Kong they are kept for the firefighters. One lift in every building is the **firefighting lift**, which only firefighters use. So almost everyone must use the stairs.

Some people cannot use the stairs on their own, for example a person in a wheelchair. In TailSafe, such a person is carried down by their family or a live-in domestic helper if someone able is at home. Otherwise they wait for the firefighters to rescue them. Rescue can take a long time, because firefighters climb the stairs and bring people down one household at a time.

Many tall buildings have refuge floors: an open floor, every so many storeys, where people can step out of the staircase into fresher air and rest before going on.

① IN PLAIN WORDS

A fire in a tall building is a slow walk down a narrow staircase, shared with hundreds of neighbours, some of whom are asleep, old or unable to walk.

CHAPTER 2

The race against smoke

Every fire is a race between two clocks: how long a place stays safe, and how long people need to leave it.

In a building fire, the smoke is often more dangerous than the flames. Smoke does two things. It makes the air **dark**, so you cannot see the way. And it is **poisonous**: breathing it makes people weak and confused, and in the end they cannot move. When a place has become too dangerous to be in, it is called **untenable**.

Fire engineers describe the race with two clocks:

- **ASET**, the *available safe egress time*: how long a place stays safe before smoke makes it untenable. Think of it as the **smoke clock**. (Egress simply means going out.)
- **RSET**, the *required safe egress time*: how long the people there need to get out. Think of it as the **walking clock**.

If the walking clock is shorter than the smoke clock, everyone leaves in time. If the walking clock is longer, someone is still there when it has become dangerous. TailSafe writes this as **RSET > ASET** (“RSET is bigger than ASET”) and counts how often it happens.



A made-up example for one floor. The floor stays safe for 18 minutes, but the last person leaves after 22 minutes. For 4 minutes someone is on a floor that is already too dangerous. That is $RSET > ASET$.

In TailSafe, a place becomes untenable at the first moment one of these is true:

- you can see **less than 10 metres** through the smoke;
- the air is **hotter than 60 °C**;

- a person who had stood there since the fire started would already have breathed a **harmful dose** of smoke. ([Chapter 11](#) explains the dose.)

These limits follow ISO 13571, an international guideline on how long people can survive in a fire. The TailSafe settings file notes that the exact values still have to be checked against the edition of the guideline in use.

IN PLAIN WORDS

ASET is how long you *have*. RSET is how long you *need*. Trouble starts when you need more time than you have.

CHAPTER 3

Why the average is not enough

This is the most important idea in TailSafe, and the reason for its name.

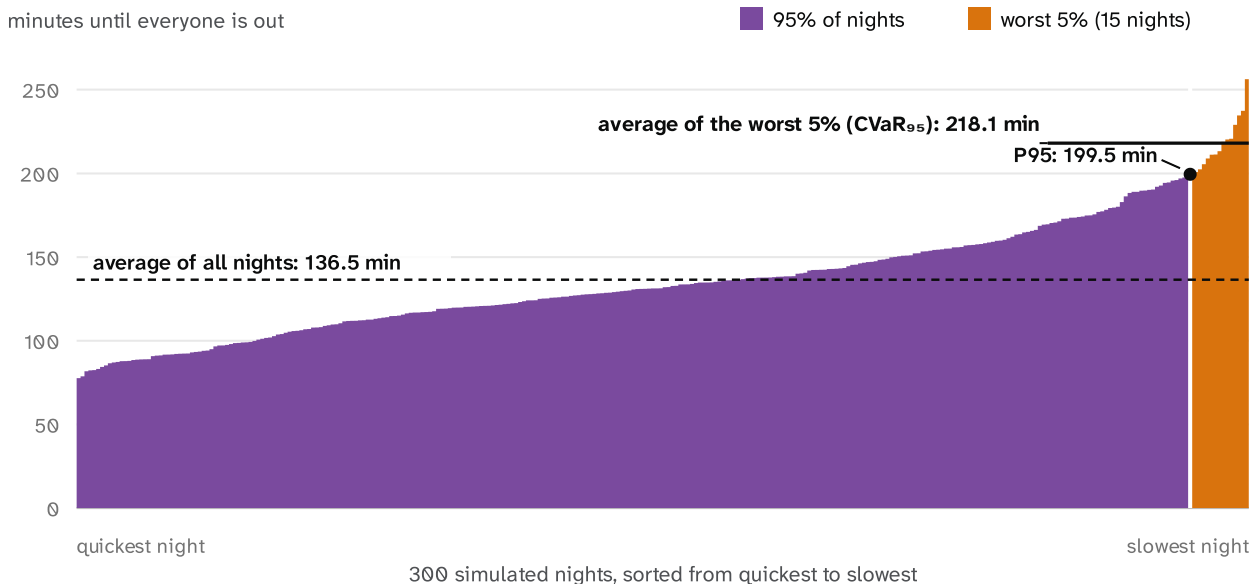
A story about buses

Think about your trip to school or work. Most days it takes 30 minutes. Some days 35. But now and then the bus breaks down and the trip takes an hour and a half. If someone asks how long your trip takes, the average (perhaps 33 minutes) is a fine answer for planning an ordinary day. But if you must never, ever be late for an important exam, the average is no use at all. You need to know about the bad days.

Fires are the same. A building checked only against an average night can look fine, while a few bad nights are very bad. And fires do not choose average nights: they happen at 3 a.m., when a staircase is full of smoke and a lift is broken.

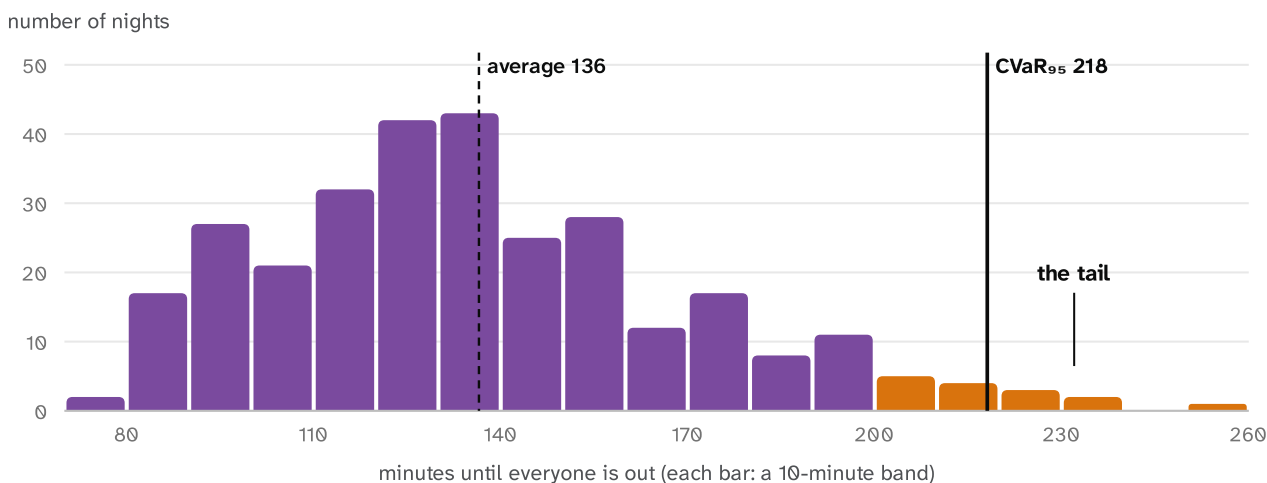
The long tail

Imagine the same fire night happening 300 times, each time a little differently: different people at home, a different flat on fire, the smoke arriving at a different moment. Draw a bar for each of the 300 nights, from the quickest evacuation to the slowest:



Each sliver is one simulated night in the 40-storey example (the reference scenario of [chapter 30](#), 300 nights, as in the browser version). The height is the time until everyone, including people rescued by firefighters, is out. Most nights end between 100 and 170 minutes. The slowest 5% of nights, in amber, are much slower.

The same 300 nights can be drawn another way, as a **histogram**: the nights are sorted into boxes by how long they took, and each bar shows how many nights landed in each box. Most nights pile up in the middle. A thin trail of slow nights stretches out to the right. That trail is called **the tail**, and it is where TailSafe gets its name.



The same 300 nights as a histogram. Bars in amber are nights slower than 200 minutes. The dashed line is the average of all nights; the solid line is the average of the worst 5%.

Four ways to describe the spread

Name	What it means	In the example
Mean (average)	Add up the times of all nights and divide by the number of nights.	136.5 min
Median	The night in the middle: half the nights are quicker, half are slower.	131.6 min

Name	What it means	In the example
P95 (95th percentile)	95 nights in every 100 are quicker than this. It is the border of the worst 5%.	199.5 min
CVaR₉₅	Take the worst 5% of nights (here the worst 15 of 300) and average them. It answers: <i>when things go badly, how badly, on average?</i>	218.1 min

CVaR₉₅ is short for *conditional value at risk at 95%*. The name comes from banks, which use it to ask “on our worst days, how much could we lose?”. TailSafe borrows the idea and asks “on the worst nights, how long does it take?”. In this guide it is often called simply **the average of the worst 5%**.

Why not simply look at the single worst night? Because one night can be a freak. The average of the worst 5% is steadier, and still tells you about the bad nights.

✓ REMEMBER

TailSafe cares most about CVaR₉₅, the average of the worst 5% of nights, because the bad nights are where people get hurt. When it tests a change, it asks first how much the change shortens the bad nights, not the average one.

CHAPTER 4

Who is most at risk

Not everyone is equally likely to be among the last people out.

- **People who are asleep** react later. At night, the time before people start to move is much longer.
- **Older residents (65 to 79)** walk more slowly, especially down many flights of stairs.
- **Frail older residents (80 and over)** walk very slowly and tire quickly.
- **Wheelchair users** cannot walk down. They must be carried, or wait for a lift or for the firefighters.
- **People on high floors** have the longest way down, and often meet queues on the way.
- **Families move together**, so a family is only as fast as its slowest member.
- **Some households first go the wrong way:** to fetch a relative or to check on a neighbour, walking against everyone else. This is called **counter-flow**.

Hong Kong's population is ageing, so the slowest groups are growing. In TailSafe, some households also have a **live-in domestic helper**, who can help an older person move or carry a wheelchair user down.

What the example shows

In the 1,000-night study of the 40-storey example ([chapter 30](#)):

- Households with a wheelchair user living on 30/F to 39/F are only **1.2%** of the residents, yet they are **all** of the people still inside at the very end of the worst nights, because they wait for the

firefighters.

- Among people who leave on their own, households with a frail resident aged 80 or over living on 30/F to 39/F are **4.9%** of the residents but **70%** of those still inside near the end of the worst nights.

IN PLAIN WORDS

The worst nights are mostly about a small number of people: the old, the frail and wheelchair users on high floors. A good plan has to work for them.

CHAPTER 5

What TailSafe is, and what it is not

TailSafe is

- a **practice tool**, like a flight simulator for evacuations;
- a safe way to ask “what if?” without a real fire or a real drill;
- a way to see **where** the risk comes from and **who** carries it;
- a fair way to **compare** changes, on exactly the same nights.

TailSafe is not

- a **fire-safety assessment** of any building;
- a replacement for a **registered fire engineer**, the **Fire Services Department**, or the Buildings Department's *Code of Practice for Fire Safety in Buildings*;
- a picture of any **real** residents: everyone in it is made up, and no personal data is used.

Many of the numbers inside TailSafe are educated guesses. TailSafe keeps every physical and population number in one settings file, `config/params.yaml`, and every number there says where it comes from. Of its 169 numbers, 146 are marked `ASSUMPTION – needs citation`: they are reasonable guesses that still need a proper source. The walking speed of frail residents, reaction times at night and how quickly firefighters rescue people are among them. The remaining numbers name a published source, which is still marked to be checked against the edition used.

That is why TailSafe is good for **understanding and comparing** (where does the risk come from? which change helps more?) and not for **exact predictions** (how many minutes will it take in my building?).

WATCH OUT

Never say “TailSafe says our building is safe”. Say instead: “TailSafe suggests where the risk concentrates and which changes might help. A fire engineer should check this.”

The full responsible-use notice, which TailSafe shows on every screen and in every report, is in [chapter 32](#).

How TailSafe works

This part opens the box. It explains what TailSafe does behind the buttons, one idea at a time. There is no maths, only pictures and everyday examples.

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The big picture

TailSafe works in seven stages. The web app follows the same order, so once you know these stages you know your way around.

1 The building Step 1

Describe the building: its floors, flats, corridors, doors, staircases and lifts.

2 The night Step 2

Describe the night you are worried about: the time, the people, the fire, the smoke, a lost staircase, a broken lift.

3 Hundreds of nights

TailSafe plays out hundreds or thousands of versions of that night, each slightly different. This is the **stress test**.

4 The tail Steps 3, 4 and 5

Look at the spread of results: how bad the worst nights are, who is still inside, where smoke catches people.

5 The cause Step 6

Find out which staircase, door or exit matters most on the bad nights.

6 The fixes Steps 7 and 8

Test cheap changes to how the building is run, and check that they really help on nights they have not seen.

7 The briefing Step 9

Write it all on one page for the building manager.

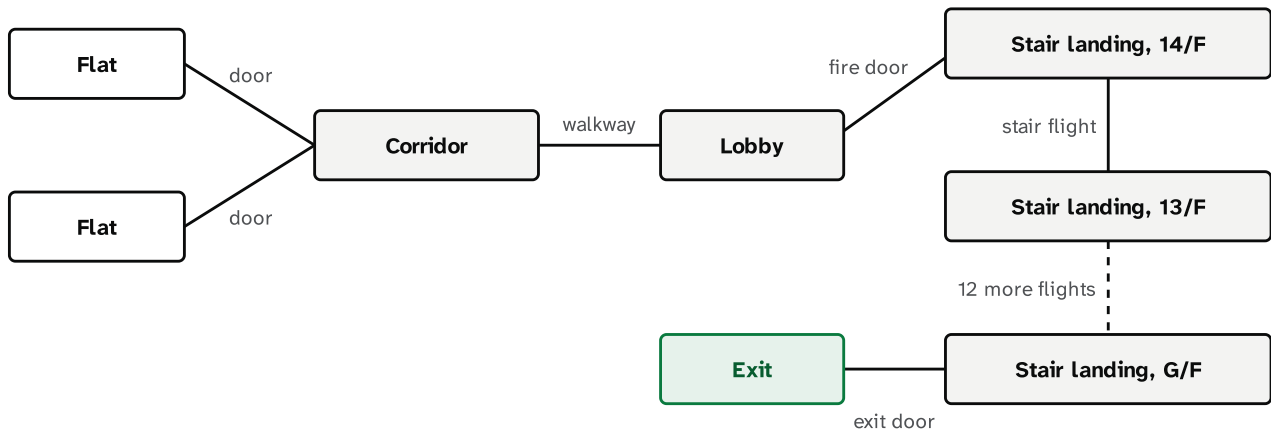
The chapters in this part follow the same stages: the building ([7](#)), the people ([8](#)), one night ([9](#)), many nights ([10](#)), the results ([11](#)), the causes ([12](#)) and the fixes ([13](#)). [Chapter 14](#) introduces four helpers that make it all easier to see and use.

The building as a map

TailSafe does not see a building as a picture. It sees a map of places, and of the connections people use to move between them.

Places and connections

The **places** are flats, corridors, lobbies, stair landings, refuge areas and exits. The **connections** are the doors, the walkways and the stair flights between them. Every connection knows how long it is and how wide it is. Mathematicians call a map like this a **graph** (nothing to do with charts): the places are its **nodes** and the connections are its **edges**.



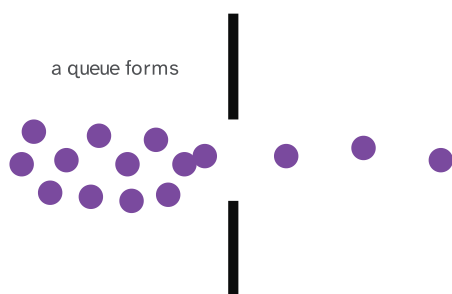
Boxes are places (nodes). Lines are connections (edges). Every connection has a length and a width.

A tiny part of the map of one floor. Real maps are much bigger: the 40-storey example has 20 flats on every typical floor, two staircases and thousands of connections.

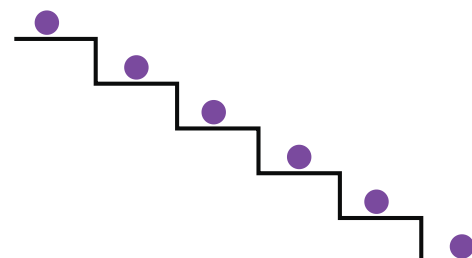
Doors and stairs are like pipes

Only so many people fit through a door or down a staircase each second, just as only so much water fits through a pipe. When more people arrive than can pass, the rest wait, and a **queue** forms, like at a supermarket checkout.

How many people can pass depends on the width. TailSafe uses a standard engineering method, the **hydraulic model** from the *SFPE Handbook of Fire Protection Engineering*. It says that a staircase lets roughly one person per second pass for each metre of **usable width**, and a door a little more. The usable width is a bit less than the real width, because people keep about 15 cm away from the wall on each side.



Door 1.0 m wide: about 55 people a minute



Stair 1.1 m wide: about 48 people a minute

The flow through a door or a stair depends on its usable width. The numbers are what TailSafe's settings give for these widths.

Here is why this matters so much. The staircases in the 40-storey example are 1.1 m wide. Take 15 cm off each side and 0.8 m is usable, which lets about 48 people a minute down. The block has about 1,825 residents at night. Just squeezing all of them through two such staircases takes about 19 minutes; through one staircase it takes about 38 minutes. So if smoke closes one staircase, the whole building has to squeeze through the other one.

Walking speed

People walk more slowly in a crowd, more slowly on stairs than on the flat, and more slowly still after many floors, because they get tired. Smoke slows them down too, because they cannot see well. TailSafe includes all four effects.

Taking turns on the stairs

On every floor, people coming out of the corridor have to squeeze into a staircase that is already carrying people from the floors above. TailSafe lets the two streams take turns in a fixed ratio, like cars merging where two lanes become one.

The four building types

TailSafe can build four kinds of Hong Kong building for you. They are **generic designs** made by the computer from a set of rules (TailSafe calls them *procedural*), not copies of real buildings.



Cruciform

A cross-shaped public housing block: four wings of flats around a central core with the lifts and two staircases. Standard size: 40 storeys.



Slab

A long, straight block: one long corridor with flats on both sides, a staircase at each end and the lifts in the middle. Standard size: 30 storeys.



Twin core

A private residential tower on a podium (a wide base without homes). Its core holds a pair of interlocking scissor staircases. Standard size: 45 storeys.



Care home

A low-rise residential care home for the elderly, where many residents need help to move and staff help them. Standard size: 4 storeys.

A **scissor staircase** is two staircases twisted around each other in the same shaft, so that each floor can reach two separate escape routes. A **storey** count includes the ground floor.

CHAPTER 8

The people

TailSafe fills the building with made-up people, called **synthetic** people. No real person's data is ever used.

Six kinds of people

Kind of person	What is special about them in TailSafe
Able adult	Walks at a normal pace and can help others.
Child	Moves together with the family.
Older adult (65 to 79)	Walks more slowly, especially down stairs, and tires sooner.
Frail older adult (80 and over)	Walks very slowly and tires quickly. Waits for an evacuation lift when lifts are in use.
Wheelchair user	Cannot walk down stairs. Is carried down by the household if an able person is at home, waits for an evacuation lift if lifts are in use, or waits for the firefighters.
Domestic helper	A live-in helper, who can help an older person move or carry a wheelchair user down.

For each kind of person TailSafe knows a walking speed on the flat, going down stairs and going up stairs, how quickly they tire, how long they usually take to react to the alarm, and how much room they take up. Speeds and reaction times are not fixed: every person gets their own, drawn from a range.

Households

Every flat gets a **household**. Its size depends on the size of the flat. The ages of its members are drawn so that, over the whole building, the share of residents aged 65 and over matches the setting you choose. Some households have a live-in helper. Some flats are empty.

Who is at home depends on the time. TailSafe knows four time slots: weekday day, weekday night, weekend day and weekend night. At night most people are at home and asleep; on a weekday afternoon many are out at work or school.

In a care home, residents live in rooms, and **staff** help them out: each member of staff escorts the residents of one room.

How people behave

- **Families stay together** and move at the pace of their slowest member.
- **Some go against the flow first:** a share of households visit another flat, on their own floor or a few floors up, to fetch family or check on a neighbour, before they leave.
- **Some rest on a refuge floor** on the way down.
- **Everyone knows the quickest way out**, and learns that a staircase is blocked only when they reach it. Then they turn around and use another one.

IN PLAIN WORDS

Every night TailSafe invents a new set of neighbours for the building, with realistic ages, speeds and habits, and then watches how they get out.

One night in the simulator

A **simulator** is a program that imitates something real. TailSafe's simulator plays out one fire night, second by second.

This is what happens during one simulated night:

- 1 **The fire starts and the alarm sounds.** Everyone in the building hears it at once. The household in the burning flat reacts quickly.
- 2 **Everyone else waits for their reaction time,** then starts to move. At night, many wait a long time.
- 3 **People move along the map.** TailSafe moves them in small time steps, like the frames of a film. At every door and staircase it checks how many people can pass in that moment; the rest queue.
- 4 **Smoke spreads from the burning flat.** It flows through open doors and into the corridors, and rises up the staircases and lift shafts, the way smoke rises up a chimney (this is called the **stack effect**). Doors that close by themselves slow it down. On some nights the burning flat's door is left open, which lets much more smoke out.
- 5 **Smoke slows people and harms them.** Where it is dark people walk more slowly, and everyone keeps a count of how much smoke they have breathed.
- 6 **A staircase may fill with smoke** at the time set in the scenario. From then on nobody can use it. People who reach it turn back and use another staircase. People already on it keep going.
- 7 **The firefighters start rescuing** people who cannot leave on their own. The time they start differs from night to night. They go to the lowest floor where someone is waiting first, climb up, and bring the household down.
- 8 **The night ends** when everyone is out. TailSafe follows the people who are moving for up to four hours after the alarm. Once only people waiting for the firefighters are left, it works out directly when each of them will be carried down, even if that is later still.

A 40-storey night takes the computer between a tenth and a third of a second, which is what makes hundreds or thousands of nights possible.

How the smoke is worked out

TailSafe's smoke model is deliberately simple. Each room, corridor, lobby and staircase is treated as one box of air, with one smoke level. Smoke flows from box to box through the openings between them, more easily upwards in stairs and lift shafts. The fire itself grows faster and faster until it reaches its full size; how fast it grows and how big it gets are rolled for every night. This is an engineering estimate, not a detailed computer fire simulation. Results from a detailed fire simulation program can be loaded into TailSafe instead.

Why it is called the “fast” simulator

The main simulator does not follow every footstep. It moves groups of people along the map's connections and keeps a queue at the end of each one, a bit like water flowing through pipes. Engineers call this a **mesoscopic** model (“meso” means “in the middle”: more detailed than a single equation, less detailed than following every step). TailSafe also has a slower, person-by-person simulator for closer looks, described in [chapter 14](#).

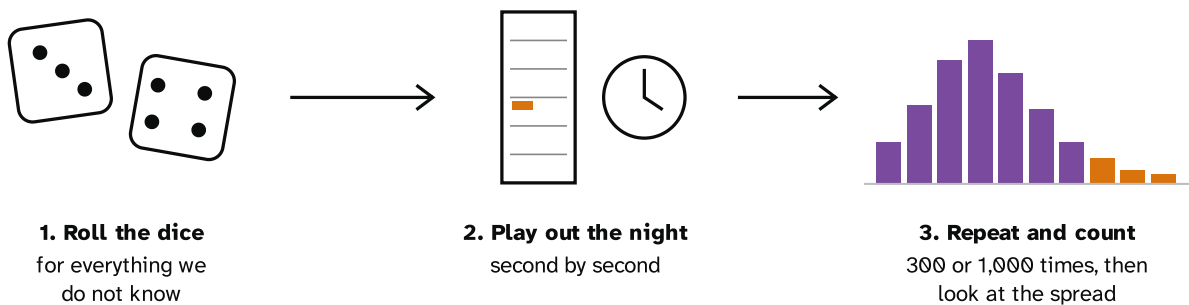
IN PLAIN WORDS

One simulated night is one possible version of the fire, played out second by second, with queues at every narrow place and smoke creeping through the building.

CHAPTER 10

Hundreds of nights

We never know exactly how a real fire night will go. So TailSafe does not play the night once: it plays it hundreds or thousands of times.



The Monte Carlo method in three steps.

Each time, TailSafe “rolls the dice” again for everything that is uncertain. This way of working things out by repeating a random experiment many times is called the **Monte Carlo method**, after the famous casino.

Rolled again every night

- who is at home, their ages, and whether a helper is there;
- everyone's walking speed and reaction time;
- which flat the fire starts in (and the floor, if you chose a random floor);
- how fast the fire grows and how big it gets;
- whether the burning flat's door is left open;

Kept the same every night

- the building;
- everything you set on the scenario screen, such as the time of day, the share of older residents, the fire floor and a staircase smoke-logged at a fixed time;
- any fix being tested, such as evacuation lifts or wardens.

- when a staircase is lost, if you gave a range of times;
- which lifts are out of service;
- when the firefighters start rescuing people.

How many nights?

You can choose 100, 300, 1,000 or 2,000 nights. More nights give more trustworthy answers but take longer. The browser version always uses 300. On a four-core computer, 1,000 nights of the 40-storey example take a little over a minute.

The seed: dice you can repeat

A computer's dice are not truly random. They come from a starting number called the **seed**. The same seed always gives exactly the same nights. That means anyone can repeat your study and get exactly the same numbers, and it means you can change one setting and be sure that everything else stayed the same.

Fair tests: the same nights for every plan

When TailSafe compares two plans, it runs both on the very same nights: the same people, the same fire, the same broken lift. Only the plan is different. It is like testing two umbrellas in the same rain, instead of one in a drizzle and the other in a storm. Statisticians call this **common random numbers**.

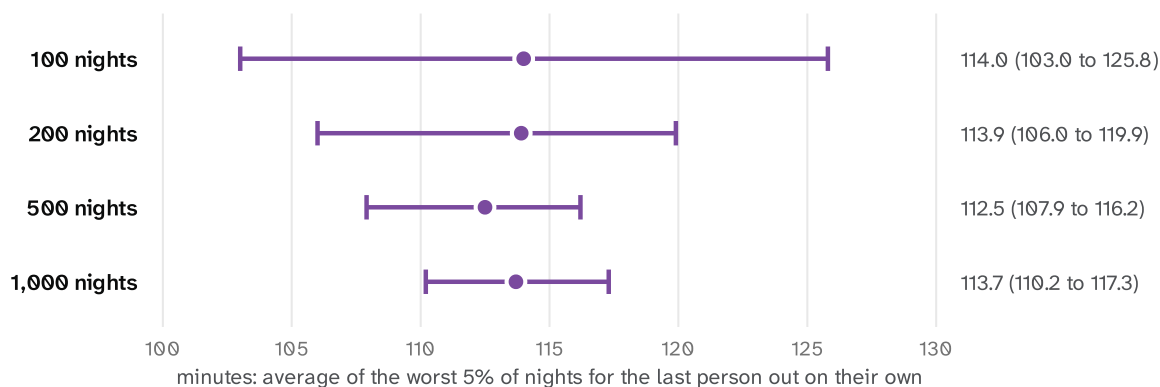


Same people, same fire, same broken lift in each pair of cards. Any difference in the result comes from the plan.

Six of the nights, tested without and with a plan. Each card shows where that night's fire started (amber) and who was at home (dots). The pairs are identical.

How sure are we? Confidence intervals

Every number worked out from a few hundred nights wobbles a little. Run another 300 nights with a different seed and you get a slightly different answer. So TailSafe gives each important number a range, the **95% confidence interval** (CI): we can be fairly sure the true value lies inside it. The more nights, the narrower the range.



The average of the worst 5% of nights for the last person out on their own, in the 40-storey example, worked out from the first 100, 200, 500 and 1,000 nights of one study. The dot is the estimate; the line is its 95% confidence interval. With 100 nights the range is about 23 minutes wide; with 1,000 nights about 7.

⚠️ WATCH OUT

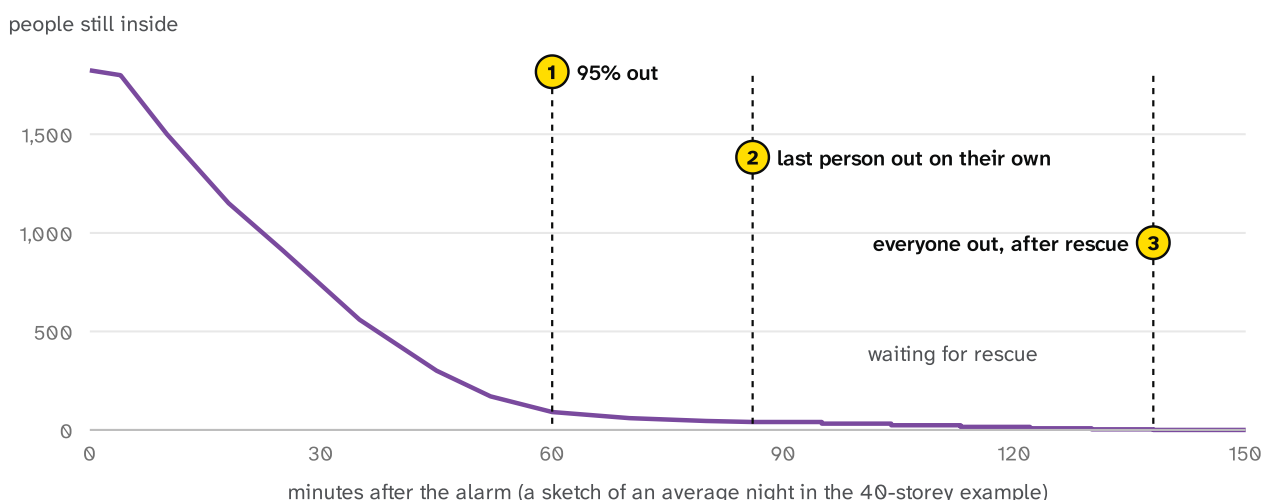
With few nights, “the worst 5%” is only a handful of nights, and their average can mislead. For the time until everyone is out in the 40-storey example, the first 100 nights gave 206 minutes and all 1,000 nights gave 219. Use at least 300 nights to explore, and 1,000 or more for comparisons that matter.

CHAPTER 11

The numbers TailSafe reports

After a stress test, TailSafe reports a handful of numbers. This chapter explains each one.

Three ways to say “how long”



A sketch of an average night: how many people are still inside as time passes. Most people are out within the first hour; a few who cannot use the stairs wait much longer for the firefighters.

On screen	What it measures
① 95% of occupants	The time until 95 out of every 100 people are out. One or two very late people do not change it, so it shows how the crowd as a whole does.
② Last self-evacuee	The time until the last person who left <i>on their own</i> is out: on foot, carried by family, or by lift. It leaves out people waiting to be rescued, so it shows how the building works for people who can move.
③ Everyone out	The time until the very last person is out, including people brought down by the firefighters. It depends heavily on how rescue works, which TailSafe can only estimate.

For each of the three, TailSafe gives the **mean**, the **P95** and the **CVaR₉₅** (see [chapter 3](#)), each with its 95% confidence interval. Read them together: “everyone out” tells you about rescue, the other two about the stairs.

The smoke dose (FED)

As a person breathes smoke, TailSafe adds up a **dose**. The dose counts mainly carbon monoxide (a poisonous gas in smoke) and heat. Fire engineers call it the **fractional effective dose**, or **FED**.

- At an FED of **1**, a person is **incapacitated**: they can no longer move by themselves.
- TailSafe treats an FED of **0.3** as already harmful, to protect the most vulnerable people. These limits follow ISO 13571.

P(RSET > ASET): how often someone is caught

This is the share of nights on which someone is still on a floor after its corridors have become untenable, or breathes a harmful dose of smoke (an FED of 0.3 or more). It is written as a number between 0 and 1: **0.60 means 60 nights in every 100**. It needs the smoke to be simulated; without smoke it shows a dash.

Who carries the tail

On each of the worst nights, TailSafe looks at the **late stragglers**: the households still inside near the end of that night. It then compares the kinds of people among the stragglers with the kinds of people in the building, using a **risk ratio**:

$$\text{risk ratio} = \text{share of the late stragglers} \div \text{share of all the people}$$

A risk ratio of 1 means a group is just as common among the late stragglers as in the building. A risk ratio of 21 means it is 21 times more common. It also names the floors where the stragglers live, for example “wheelchair users on 30/F to 39/F”.

Floors that fail, and where the stairs queue

Floors that fail shows, for each floor, how often someone is still there after it has become untenable. Floors at and just above the fire are usually the worst. **Where stairs queue** shows how much queueing happens on each flight of each staircase on the worst nights, in **person-minutes**: if 10

people wait for 3 minutes, that is 30 person-minutes.

CHAPTER 12

Finding the cause

A **bottleneck** is a narrow place that slows everything behind it, like the neck of a bottle. TailSafe finds the bottlenecks that make the worst nights bad.

The obvious approach is to look for where the crowd is thickest. TailSafe goes one step further and asks a **what-if** question for every candidate: *if this part of the building were better, how much shorter would the worst nights be?* Asking what would have happened if something had been different is called a **counterfactual** test.

The candidates are:

- each whole staircase, all its flights at once;
- the doors of each staircase;
- each final exit;
- the places where the longest queues form on the worst nights;
- any staircase the scenario blocks with smoke. Here the test is “what if it had stayed usable?”.

For each candidate, TailSafe gives it more room (50% more capacity by default; you can also choose 25% or 100%), re-runs the bad nights with exactly the same dice, and measures how much the average of the worst 5% falls. The candidates are then ranked, the biggest improvement first, each with its confidence interval.

TailSafe also works out two simpler facts about the building's escape routes:

- **Maximum egress flow:** the most people per second the whole building could possibly get out, if everyone were already queued at the right places. In the 40-storey example it is 1.62 people per second, which is exactly two staircases' worth.
- **Minimum cut:** the connections that set that limit, the narrowest point of the whole system. In the example it is the bottom flight of each staircase.

IN PLAIN WORDS

TailSafe does not guess which door matters. It tries improving each one on the bad nights and measures the difference.

Testing fixes

TailSafe looks for **operational** fixes: changes to how a building is run, not to its walls. They are cheap and quick compared with building work.

The five kinds of fix

Fix	What it means	The catch
Evacuation lifts	Use the ordinary (not the firefighting) lifts to carry residents who cannot manage the stairs: wheelchair users and frail older adults, or wheelchair users only. The lifts start after a short changeover and can serve the highest floors first, the nearest waiting floor, or the lowest floors first.	People who wait for a lift may get out later than if they had walked.
Hold stair doors open	Keep the staircase doors open, so more people can pass.	Smoke also gets into the staircases more easily.
Stair assignment by floor	Tell some floors to use one staircase and the rest the other, so the streams do not collide. If the assigned staircase is blocked, people use whatever is quickest.	A staircase can end up with too many people.
Phased release	Let some floors leave first while others wait in their flats for a set time.	People waiting in their flats still breathe any smoke that reaches them.
Floor wardens	Trained residents or staff on a floor knock on every door of their floor and the floors next to it, so people react sooner, and each helps one household that cannot use the stairs alone.	They have to be recruited and trained.

How TailSafe searches

There are far too many combinations to try them all, so TailSafe searches step by step, testing every candidate on the same set of nights:

1

Try each fix on its own

For example, lifts with each way of choosing floors, or a warden on each likely floor.

2

Combine the good ones

Start from the best single fix and add the others one at a time, as long as the result keeps getting better.

3

Add wardens

One at a time, up to the number you allow.

4

Fine-tune the timing

If letting floors leave in phases helped, adjust the waiting times. (The method used is called CMA-ES.)

5

Confirm on fresh nights

Test the winning plan against doing nothing, on new nights it has never seen, and report only this result.

Why the fresh nights?

If you try 30 plans on the same 60 nights, the plan that looks best is partly lucky: it happens to suit those particular nights. This is called the **winner's curse**. It is like a student who memorised last year's exam: the real test is a new exam. So TailSafe tests the winning plan again on fresh nights (200 by default) and only then says whether it helps.

Better, worse, or no clear change

For every outcome, TailSafe shows the value without the plan, with the plan, the change, and the confidence interval of the change. Then it gives a verdict:

- **better**: the whole range of the change is below zero;
- **worse**: the whole range is above zero;
- **no clear change**: the range includes zero, so we cannot tell.

✓ **REMEMBER**

A fix can make one thing better and another worse. In the 40-storey example, evacuation lifts halve the time until everyone is out, but make the time for 95% of people to get out several minutes longer. TailSafe always shows every outcome and warns you about trade-offs. It never hides them.

CHAPTER 14**Four helpers**

Four more parts of TailSafe make the results easier to see, quicker to explore and easier to share.

1. The person-by-person replay

The fast simulator treats crowds like water in pipes. For a closer look, TailSafe has a second, **microscopic** simulator that moves every single person as a small disc across the floor plan, keeping a safe gap to whoever is in front (a *collision-free speed model*, after Tordeux, Chraibi and Seyfried, 2016). It is much slower, so TailSafe uses it only to replay single nights, and to check that the two simulators roughly agree. In a 20-storey block they agree on when the last walker leaves to within a few seconds on average; for most of the crowd, the detailed one is 7 to 15% slower, because real queues at doors take up room.

2. The quick guesser

A **neural network** is a program that learns patterns from examples. TailSafe's network has studied 800 simulated buildings and nights. Show it a building and some settings, and it estimates the results in a fraction of a second, instead of a minute or more. It passes messages along the building's map, from flat to corridor to staircase, which is why it is called a **graph neural network**. Its formal name is a **surrogate**: a stand-in for the simulator.

On buildings like the ones it learned from, its estimates of the bad nights are usually within 10 to 20% of the simulator. On a kind of building it has never seen, it can be off by a third or more.

WATCH OUT

The quick guesser learned from the simulator, not from real fires, so it can only be as good as the simulator, and usually a little worse. It warns you when your settings are outside what it has learned. Before you rely on a number, press **Confirm with full simulation**.

3. The briefing

A one-page summary for a building manager, which can be saved as a PDF. TailSafe writes it by filling fixed sentences with the numbers from your results, and then checks every number in the text against the results. Whoever runs the full app can also switch on an optional AI writer (a large language model) to draft the text. It is off unless it is set up. If its draft contains any number that is not in the results, the draft is thrown away and the fixed-sentence briefing is shown instead.

4. The floor-plan reader

It turns a picture of a floor plan into a TailSafe building. It finds the walls, rooms, doorways and staircases in the picture; you correct its mistakes; then it stacks the floor into a tower. It works in the full app only. [Chapter 29](#) shows how to use it.

Getting TailSafe running

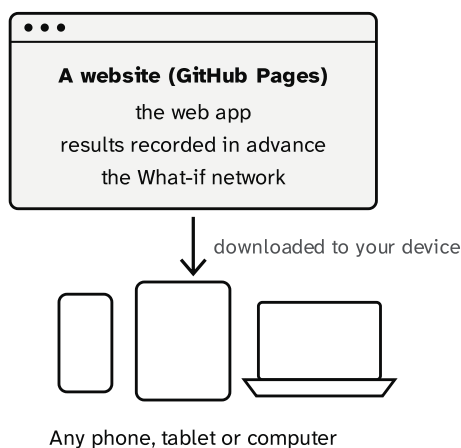
There are two ways to use TailSafe: the browser version, which anyone can open in a few seconds, and the full app, which runs on your own computer and can study any building and any night. This part explains both, click by click.

15	Two ways to use TailSafe	27
16	The browser version, step by step	28
17	The full app with Docker, step by step	31
18	Other ways to run the full app	33

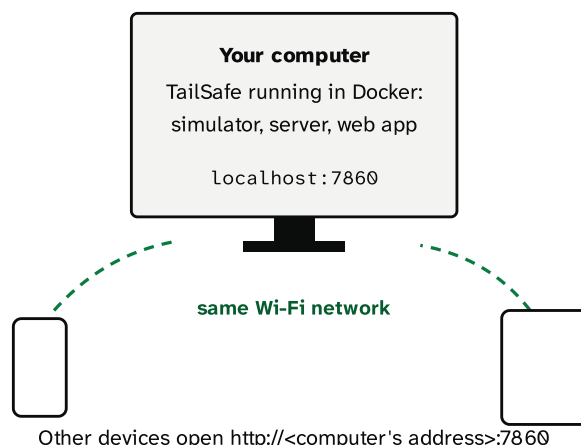
Two ways to use TailSafe

The **browser version** is a website: nothing to install, works on any phone. The **full app** runs on your own computer and can simulate anything you ask.

Browser version



Full app



Left: the browser version is a set of files that your device downloads from a website; the quick What-if network then runs on your device. Right: the full app runs on one computer, and other devices on the same network can use it through their web browser.

	Browser version	Full app
What you need	A web browser	A computer with Docker (or Python and Node)
Setting up	None: open a web address	A one-off build of several minutes
Works on	Any phone, tablet or computer	The computer running it, and any device that can reach it
Buildings	The four standard building types, at their standard sizes	The four types at any size, a floor-plan image, or your own building file
Scenarios	The reference scenario of each building (Sunday, 3 a.m.)	Any settings you like
Stress test, 3D view, replay, bottlenecks, plan, briefing	Recorded in advance from the simulator, 300 nights per building	Worked out when you ask, 100 to 2,000 nights
What-if	Yes, for any settings (it runs on your device)	Yes, plus Confirm with full simulation for any settings
Floor-plan reader	No	Yes
Where the work happens	On your device; nothing you enter is uploaded	On the computer running the full app

Which one should I use?

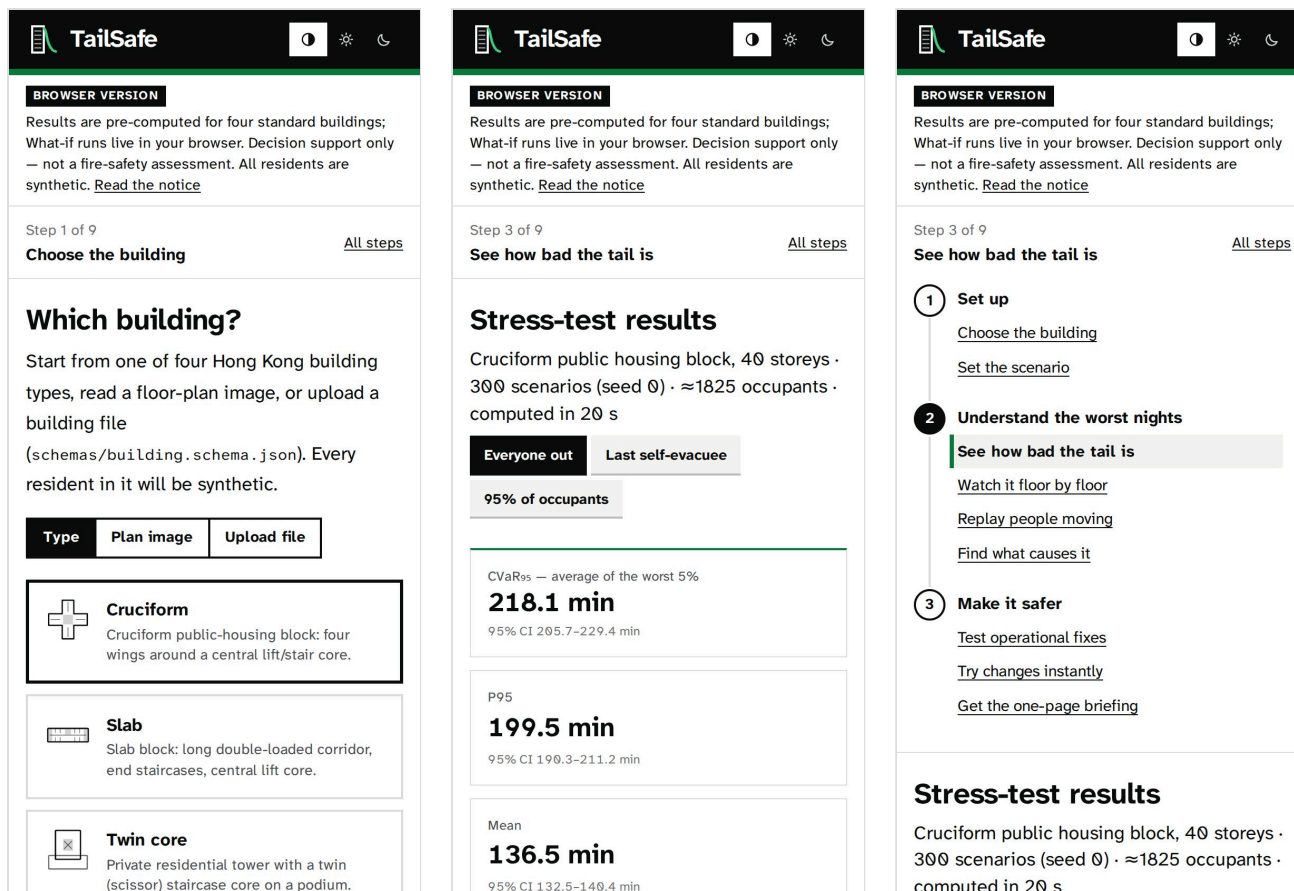
If you want to...	Use
See what TailSafe does, or show it to someone	the browser version
Look at the four example buildings on your phone	the browser version
Get a quick estimate for different settings	the What-if screen, in either version
Run a full stress test with your own settings, such as a daytime fire or more older residents	the full app
Study a building of a different size, your own building file, or a floor-plan image	the full app
Let a team use it from their own phones and laptops	the full app on one computer they can reach, or on a public web address (chapter 18)

CHAPTER 16

The browser version, step by step

You need a phone, tablet or computer with an up-to-date web browser (such as Chrome, Edge, Firefox or Safari) and an internet connection. That is all.

- 1 Open your web browser.**
- 2 Go to this address:**
`https://chingkheinganba231005.github.io/Tail_Safe/`
Type it into the address bar at the top of the browser and press Enter (on a phone, tap Go).
- 3 Wait a moment** for the page to load. You will see the black TailSafe bar at the top and, just below it, a small black label saying **BROWSER VERSION**.
- 4 Follow the steps.** On a computer they are listed down the left side. On a phone, tap **All steps** near the top to see them. Part 4 of this guide explains every step.



The browser version on a phone. Left: step 1. Middle: the results of the stress test. Right: the list of steps, opened with **All steps**.

What you can do

- Use all nine screens for the four standard buildings.
- See the results, the 3D view, the person-by-person replay, the bottleneck ranking, the recommended plan and the briefing, all recorded from the simulator.
- Use the What-if screen with *any* settings: the quick guesser runs on your device.
- Download the one-page briefing as a PDF.

If you change a setting

The browser version only holds results for the reference scenario of each building. If you change the scenario, the **Run stress test** button stops working and this message appears. Press **Back to the reference scenario** to undo your changes, or use the What-if screen to get a quick estimate for your settings.

What needs the full app

- Changing the size of a building, reading a floor plan, or uploading your own building.
- A full stress test with settings other than the reference scenario.
- Replays of nights other than the worst one and the median one.
- Ranking bottlenecks or searching for a plan with other options.

① You have changed the scenario. The browser version only has results for the reference scenario of each building — the What-if screen estimates any settings instantly, in your browser.

[Back to the reference scenario](#)

The message shown when the scenario differs from the recorded one.

Good to know

- **Nothing you enter is uploaded.** The browser version only downloads pages and results recorded in advance; there is no server that stores or works on what you type.
- **Nothing is saved.** If you reload the page, you start again at step 1. Only your choice of light or dark colours is remembered.
- **Every chart can be read as a table.** Use the **Chart / Table** switch above it. On a phone, turning the phone sideways gives charts more room.
- **The 3D views need WebGL**, a graphics feature built into almost every modern browser. If a 3D view stays blank, see [chapter 33](#).
- **Tap “Read the notice”** under the black bar to read the full responsible-use notice.

The full app with Docker, step by step

Docker is a free program that runs software inside a **container**: a sealed box that holds everything the software needs. With Docker you do not have to install Python or anything else yourself.

Step 1. Install Docker (once)

- **Windows or Mac:** download **Docker Desktop** from <https://www.docker.com/products/docker-desktop/> and install it. On a Mac, choose the version for your chip (Apple or Intel). The installer may ask you to restart the computer. Then open Docker Desktop and wait until it says that Docker is running.
- **Linux:** install **Docker Engine** by following the steps for your system at <https://docs.docker.com/engine/install/>.

Step 2. Download TailSafe (once)

- **If you know git:**

```
git clone https://github.com/chingkheinganba231005/Tail_Safe.git
```

- **If you do not:** open https://github.com/chingkheinganba231005/Tail_Safe, click the green **Code** button, then **Download ZIP**. Unzip the file. You get a folder called `Tail_Safe-main`.

Step 3. Open a terminal in the TailSafe folder

A **terminal** is a window where you type commands.

- **Windows:** open the TailSafe folder in File Explorer, click the address bar at the top, type `powershell` and press Enter.
- **Mac:** open *Terminal* (in Applications, then Utilities), type `cd` followed by a space, drag the TailSafe folder into the window, and press Enter.
- **Linux:** right-click inside the folder and choose *Open in Terminal*.

Step 4. Build TailSafe (once, and after each update)

```
docker build -t tailsafe .
```

Type it exactly, including the full stop at the end, which means “this folder”. The first build downloads a lot of software and can take several minutes. Later builds are quicker, because Docker remembers the parts that have not changed.

Step 5. Start TailSafe

```
docker run -p 7860:7860 tailsafe
```

After a few seconds a line like `Uvicorn running on http://0.0.0.0:7860` appears. TailSafe is now running. Leave this window open.

Step 6. Open it

In your web browser go to `http://localhost:7860`. **localhost** means “this computer”, and **7860** is the **port**: a numbered door on your computer where TailSafe is listening. The label under the black bar now says **PROTOTYPE** instead of BROWSER VERSION, and every option works.

Step 7 (optional). Use it from a phone or tablet

- 1 Make sure the phone and the computer are on the **same Wi-Fi network**.
- 2 Find the computer's **IP address** (its address on your network), for example `192.168.1.23`.
 - Windows: in PowerShell type `ipconfig` and look for *IPv4 Address*.
 - Mac: *System Settings*, then *Wi-Fi*, then *Details* next to your network.
 - Linux: type `hostname -I` and use the first address.
- 3 On the phone, open `http://192.168.1.23:7860`, using your computer's address.
- 4 If it does not load, the computer's firewall may be blocking it. Allow Docker through the firewall when asked, and check the address again.

Step 8. Stop it

Press **Ctrl** and **C** together in the terminal window, or stop the container in Docker Desktop. To start again later, repeat step 5; you do not need to build again.

① GOOD TO KNOW

- **The first simulation is slower.** After each start, the simulator needs about ten seconds to prepare itself the first time it runs.
- **How long runs take.** On a four-core computer, 300 nights of the 40-storey example take about 20 seconds and 1,000 nights a little over a minute. Ranking bottlenecks and searching for a plan re-run many nights, so they take several minutes.
- **Repeats are instant.** While TailSafe is running it remembers results, so asking for the same thing twice gives the answer at once. When you stop the container, it forgets.
- **To update TailSafe,** download it again (or run `git pull` in the folder), then build again (step 4).

Other ways to run the full app

From the source code (for developers)

If you want to change TailSafe itself, run it directly from its code. You need **Python 3.11** or newer, **Node 20** or newer, **git** and **make**. These commands are for macOS and Linux; on Windows, use Docker or the Windows Subsystem for Linux (WSL).

```
git clone https://github.com/chingkheinganba231005/Tail_Safe.git
cd Tail_Safe
make install    # set up Python and install TailSafe (once)
make web        # build the web app (once, and after changing it)
make api        # start TailSafe
```

Then open `http://localhost:8000`. While changing the web app itself, `make dev` starts TailSafe with a web app that reloads whenever you save a file, at `http://localhost:5173`. `make check` runs all the tests. The development guide, `DEVELOPMENT.md`, has the details.

At a public web address (Hugging Face Space)

Hugging Face is a website that can run a Docker container for you, for free, at a public web address; such a hosted app is called a **Space**. TailSafe comes with an automatic workflow that sends the full app to your Space whenever the code on GitHub changes:

- 1 Create a free account on `huggingface.co`.
- 2 Create a new **Space** and choose **Docker** as its type. The free CPU hardware is enough. Note its name, for example `yourname/tailsafe`.
- 3 In your Hugging Face settings, create an **access token** with *write* permission, and copy it.
- 4 On GitHub, open the TailSafe repository, then *Settings, Secrets and variables, Actions*. Add a **secret** called `HF_TOKEN` containing the token, and a **variable** called `HF_SPACE` containing the Space name.
- 5 Open the *Actions* tab, choose **Full app (Hugging Face Space)** and press **Run workflow**. When the Space has finished building, open it at `https://huggingface.co/spaces/yourname/tailsafe`.

WATCH OUT

Anyone with the address can use it, and what they enter goes to that server. The free hardware has two processor cores, so runs take about twice as long as on a four-core laptop, and only one simulation runs at a time. A free Space may pause when nobody has used it for a while; the next visitor then waits for it to start.

Screen by screen

TailSafe has nine screens, one for each step. This part walks through every one: what it is for, what to press, and how to read what you see. The pictures come from the browser version, using the 40-storey example; the full app looks the same.

19	Finding your way around	35
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21	Step 2: Set the scenario	39
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23	Step 4: Watch it floor by floor	45
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Finding your way around

Every screen has the same frame around it. Once you know the frame, you always know where you are and where to go next.

The screenshot shows the TailSafe web application interface. At the top is a black bar with the TailSafe logo (1) and a settings icon (2). Below this is a green bar with a 'BROWSER VERSION' notice (3). The main content area is titled 'Which building?' (5) and includes a sidebar with a step list (4). The sidebar has three sections: 'Set up' (1), 'Understand the worst nights' (2), and 'Make it safer' (3). The 'Which building?' section (5) has a title 'Which building?' and a description: 'Start from one of four Hong Kong building types, read a floor-plan image, or upload a building file (schemas/building.schema.json). Every resident in it will be synthetic.' Below this are three tabs: 'Building type' (selected), 'Floor-plan image', and 'Upload file'. Under 'Building type' (6), there are four options: 'Cruciform' (with a cross icon), 'Slab' (with a grid icon), 'Twin core' (with a scissor icon), and 'Care home' (with a building icon). Each option has a brief description. Below these are input fields for 'Storeys' (40), 'Flats per wing' (5), 'Lifts' (3), 'Flat width' (6), and 'Flat depth' (8). There is a checkbox for 'Wing end stairs' and a green 'Generate building' button (7). At the bottom, there are three buttons: 'Hundreds of nights, not one', 'Look hard at the worst 5%', and 'Then test cheap fixes'.

The first screen on a computer, with its parts numbered.

- 1 **The black bar.** Click the TailSafe name at any time to go back to step 1.
- 2 **Colour buttons:** match your device, always light, or always dark. TailSafe remembers your choice.
- 3 **The notice.** The label shows which version you are using (**BROWSER VERSION** or, in the full app, **PROTOTYPE**) and reminds you that TailSafe is decision support only. Click *Read the notice* for the full text.
- 4 **The step list:** nine steps in three groups. Click any step to go to it. Steps you cannot use yet are greyed out: most need a building first, and the results need a stress test first.
- 5 **Where you are:** “Step 1 of 9”.
- 6 **The screen itself.** Its title says what it is for.
- 7 **Green buttons** do the main thing on each screen, such as **Generate building** or **Run stress test**.

Back and Continue

At the bottom of every screen there is a **Back** link to the previous step, and from step 3 onwards a green **Continue** button to the next one. Steps 1 and 2 move on with their own buttons, **Confirm building** → and **Run stress test**. So you can go through all of TailSafe by pressing the green buttons, one after another.

The footer at the very bottom of every screen links to the project page, this guide, the assumptions, the validation and the source code.

On a phone

On a narrow screen the step list is folded away. Near the top you see “Step 3 of 9” and the name of the step; tap **All steps** to open the list (see the phone pictures in [chapter 16](#)). Everything else works the same way.

Reading the charts

- **Every chart has a table.** Use the **Chart / Table** switch above it to see the exact numbers.
- **Point at a bar or a line** (or tap it on a phone) to see its value.
- **Colours.** Purple, orange and green show data. Warnings always come with a symbol and words, never with colour alone.
- **Waiting.** Long jobs show a progress bar with the number of nights done. In the browser version answers appear almost at once, because they were recorded in advance.

Step 1: Choose the building

Pick the building to test. Everything that follows is worked out for this building.

Step 1 of 9

Which building?

Start from one of four Hong Kong building types, read a floor-plan image or upload a building file (schemas/building.schema.json). Every resident in it will be synthetic.

Building type

Floor-plan image

Upload file

Cruciform

Cruciform public-housing block; four wings around a central lift/stair core.

Slab

Slab block: long double-loaded corridor, end staircases, central lift core.

Twin core

Private residential tower with a twin (scissor) staircase core on a podium.

Care home

Low-rise residential care home for the elderly (high-dependency occupants).

Stores

Flats per wing

Lifts

☐ Wing end stairs

Flat width

Flat depth

40

5

3

6

8

Generate building

In the browser version each building type has its standard size, and reading floor plans needs the full app, which can change both.

Stores

Flats / rooms

Staircases

Lifts

Height

40

760

2

3

114 m

Stair A, Stair B

Lift 1, Lift 2, Lift 3 (firefighting)

Cruciform public housing block, 40 storeys

20 flats per typical floor on four double-loaded wings around a central core with two staircases and 3 lift(s).

Floor: 1/F

Grey: circulation and stair landings (labelled). Dark lines: doors. Green dots: final exits.

Check and correct

Clear widths in metres. Changes are validated by the server.

Staircases

Stair A (dogleg) 1.1

Stair B (dogleg) 1.1

Final exits

Main entrance (N) door 1.6

Main entrance (S) door 1.6

Stair A discharge door 1.6

Stair B discharge door 1.6

Apply corrections

Confirm building →

Step 1 after pressing **Generate building**.

Do this

- 1 Make sure **Building type** is selected, then click one of the four building cards.
- 2 *Full app only*: change the options if you want a different size. The table below lists them.
- 3 Click **Generate building**. The numbers, the plan and the correction panel appear.
- 4 Look through a few floors with the **Floor** menu. Floors marked (*refuge*) or (*ground*) differ from the typical floor.

- 1 **Where the building comes from:** **Building type**, **Floor-plan image** (chapter 29) or **Upload file**. The last two need the full app.
- 2 **The four building types**. Click one; a black border shows your choice.
- 3 **Size options**, such as the number of storeys. Fixed in the browser version.
- 4 **Generate building** makes the building.
- 5 **The building in numbers**: storeys, flats (or rooms), staircases, lifts (the firefighting lift is marked) and height.
- 6 **The floor plan**. Choose any floor from the **Floor** menu.
- 7 **Check and correct**: the clear width of each staircase and each final exit, in metres.
- 8 **Confirm building** → takes you to step 2.

- 5 *Full app only:* if a staircase or exit is wider or narrower in your building, type its clear width and click **Apply corrections**. TailSafe checks the change before using it.
- 6 Click **Confirm building** →.

Reading the plan

Grey areas are corridors, lobbies and stair landings; each staircase is labelled with its letter (A, B, and so on). Dark lines across openings are doors. Green dots are the final exits to the outside. White boxes are flats (or, in a care home, rooms).

The size options

Building type	Options (standard values in brackets)
Cruciform	Storeys (40), flats per wing (5), lifts (3), wing end stairs (off: when on, adds a staircase at the end of each of the four wings), flat width (6 m) and depth (8 m).
Slab	Storeys (30), flats per side of the corridor (10), lifts (2), flat width (6 m) and depth (8 m).
Twin core	Storeys (45), podium levels without homes (3), flats per floor (8), lifts (3), protected lobbies in front of the staircases (on), flat width (7 m) and depth (9 m).
Care home	Storeys (4), rooms per side of the corridor (6), lifts (1), room width (4.5 m) and depth (6 m).

▷ TRY IT (FULL APP)

Generate the cruciform block twice, once with 20 storeys and once with 40, and run the same stress test on both (step 2). How much longer do the worst nights take in the taller block?

Upload file (full app) takes a building described in a JSON file that follows TailSafe's building schema (`schemas/building.schema.json`). The easiest way to make one is to start from a generated building: `tailsafe building generate` writes such a file ([chapter 31](#)).

Step 2: Set the scenario

A **scenario** describes the fire night you want to test. You set what you know; everything else is rolled again for every night.

Step 2 of 9

Scenario

Fix what is known; everything else (who is home, reaction times, fire growth, when the fire service arrives...) is sampled from the parameter registry in every run.

Load the reference scenario

1

When and who

Weekday day

Weekday night

Weekend day

Weekend night (3 a.m.)

Residents aged 65+ ☐ registry default 22%

Of those, aged 80+ ☒ registry default

Vacant flats ☒ registry default

Households that first go against the flow (fetch family, check on neighbours) ☒ registry default

3 Fire and smoke

☒ Simulate smoke spread and tenability (needed for $P(RSET > ASET)$)

Fire floor

14/F

Fire flat door left open

☒ registry default

Staircases

☒ Stair A smoke-logged

at

4

min

☐ Stair B smoke-logged

Chance that one more random staircase is lost

0%

5 Lifts and rescue

Lifts out of service (chosen at random each run)

1

☐ Use non-firefighting lifts to evacuate mobility-impaired residents

Fire-service rescue teams

Empty = registry default

Monte Carlo runs

300

Seed

0

Run stress test

7

Blocked: Stair A at 4.0 min. Same seed → same scenarios, so runs are reproducible and comparable.

← [Back: choose the building](#)

The scenario screen with the reference scenario: a Sunday at 3 a.m., 22% of residents aged 65 or over, fire on 14/F, Stair A smoke-logged after 4 minutes, one lift out of service.

- 1 **Load the reference scenario** puts back the standard Sunday, 3 a.m. night.
- 2 **When and who:** time of day and the mix of residents.
- 3 **Fire and smoke:** whether smoke is simulated, the fire floor, and how often the burning flat's door is left open.
- 4 **Staircases:** which staircase fills with smoke, and when.
- 5 **Lifts and rescue:** broken lifts, evacuation lifts and fire-service rescue teams.
- 6 **How many nights** to simulate, and the **seed** ([chapter 10](#)).
- 7 **Run stress test** starts the simulation.

Every setting explained

Setting	What it does
Time of day	Weekday day, weekday night, weekend day, or weekend night (3 a.m.). It changes who is at home and how quickly they react.
Residents aged 65+	The share of residents aged 65 or over (up to 60%). Untick <i>registry default</i> to set your own.
Of those, aged 80+	Of the older residents, the share who are 80 or over (the frail group).
Vacant flats	The share of empty flats (up to 50%).
Households that first go against the flow	The share of households that first fetch family or check on a neighbour.
Simulate smoke spread and tenability	Switches the smoke model on. Needed for $P(RSET > ASET)$ and for smoke to slow and harm people.
Fire floor	A chosen floor, or <i>Random floor</i> (a new floor every night).
Fire flat door left open	How often the burning flat's door is left open (only with smoke on).
Stair A smoke-logged (one line per staircase)	Tick to lose that staircase. Choose <i>at</i> and a time in minutes, or <i>between</i> two times for a different random time every night.
Chance that one more random staircase is lost	On this share of nights, one more staircase, chosen at random, is lost at a random time in the first 15 minutes.
Lifts out of service	How many lifts are broken. Which ones is chosen at random every night.
Use non-firefighting lifts to evacuate mobility-impaired residents	Turns on evacuation lifts, with Dispatch (highest floors first, nearest waiting floor, or lowest floors first) and Who may wait for a lift (wheelchair users and frail older adults, or wheelchair users only).
Fire-service rescue teams	How many rescue teams the firefighters use. Leave it empty for the standard number.
Monte Carlo runs	How many nights: 100, 300, 1,000 or 2,000.
Seed	The starting number for the dice. Same seed, same nights.

“Registry default” means TailSafe uses its standard value from its settings file (the *parameter registry*, `config/params.yaml`).

Do this

- 1 For a first look, keep the reference scenario that is already loaded.
- 2 *Full app only*: change the settings you want to test, and choose the number of nights (300 to explore).
- 3 Click **Run stress test**. A progress bar counts the nights. When it finishes, TailSafe opens step 3 by itself.

▷ TRY IT (FULL APP)

Run the reference scenario with 300 nights. Then untick **Stair A smoke-logged** and run again with the *same seed*. Because the seed is the same, the people and the fires are the same, so any difference is caused by the staircase alone.

⚠ WATCH OUT

In the browser version, **Run stress test** only works for the reference scenario. If you change anything, a message offers to put it back ([chapter 16](#)). The What-if screen (step 8) can estimate any settings.

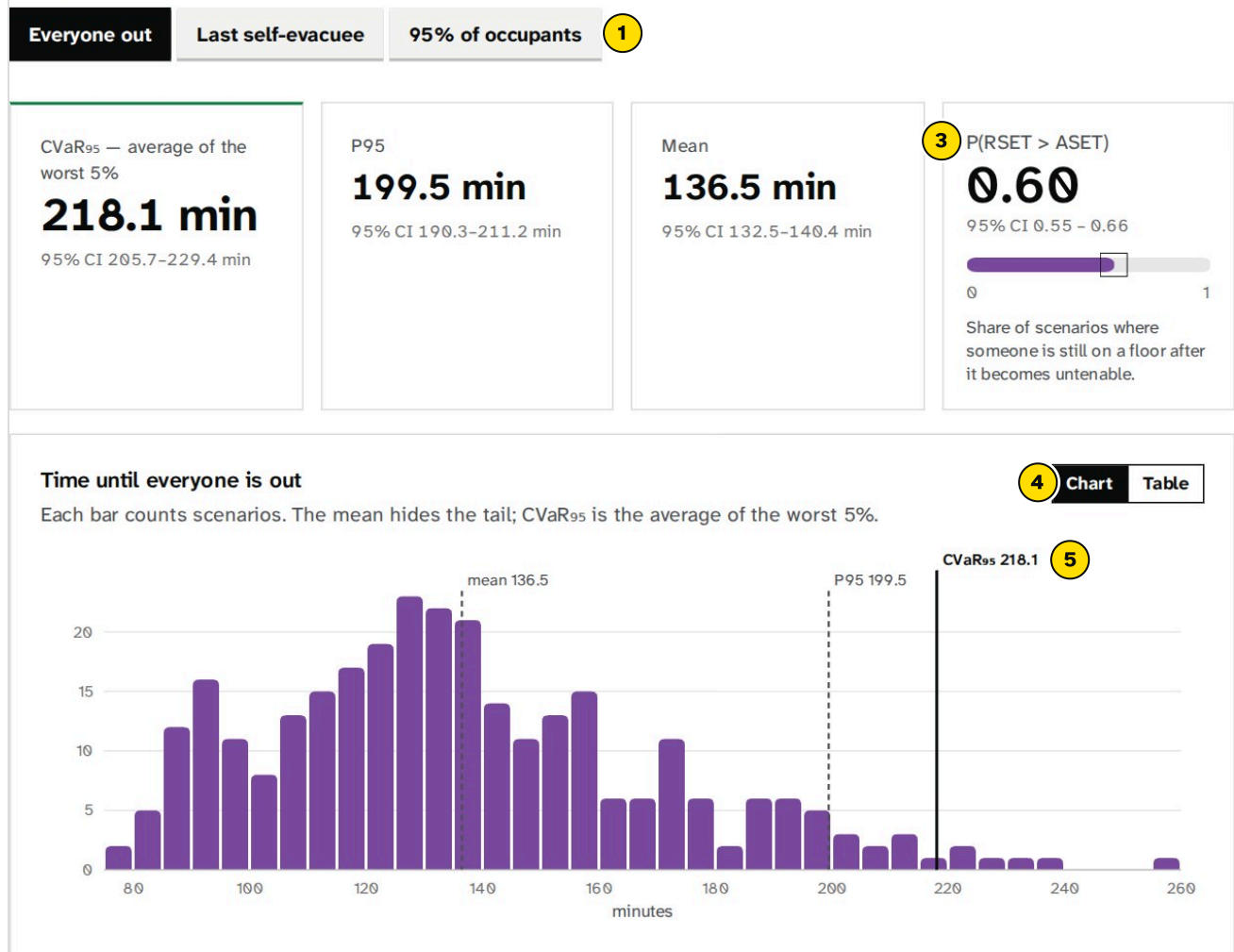
Step 3: See how bad the tail is

This is the heart of TailSafe: how long the evacuation takes over all the simulated nights, and how bad the worst nights are.

Step 3 of 9

Stress-test results

Cruciform public housing block, 40 storeys · 300 scenarios (seed 0) · ≈1825 occupants · computed in 20 s



The top of the results screen for the 40-storey example (300 nights).

- 1 **Which outcome** to show: **Everyone out**, **Last self-evacuee** or **95% of occupants** (chapter 11). Everything below changes with it.
- 2 **CVaR₉₅**, the **average of the worst 5%**. The most important number, marked with a green line. Under it, its 95% confidence interval. Next to it: **P95** and the **mean**, each with its interval.
- 3 **P(RSET > ASET)**: how often someone is caught by smoke, as a bar from 0 to 1 with its interval. Here 0.60: on 60 nights in 100.
- 4 **Chart / Table** switch.

- 5 **The histogram:** each bar counts nights. Dashed lines mark the mean and P95; the solid line marks $CVaR_{95}$. The long stretch on the right is the tail.

The line under the title tells you what the numbers are based on: the building, the number of nights and the seed, about how many people were in the building, and how long the computer took.

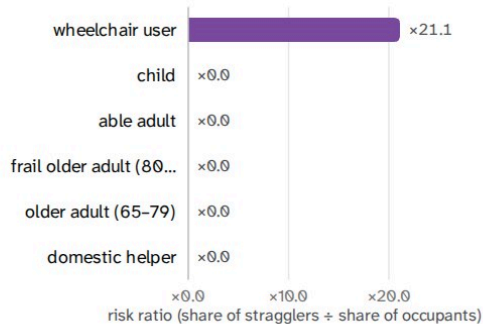
Further down the screen

Who is still inside in the worst 5%

Households whose most dependent member is a wheelchair user, on 30/F–39/F, are 1.2% of occupants but 100% of the people still inside late in the worst 5% of scenarios.

Chart

Table



Who is still inside in the worst 5%. The sentence at the top gives the headline. The bars show the risk ratio of each kind of household: here, households with a wheelchair user are 21 times more common among the last people out than in the building.

Floors that fail

$P(\text{someone is still on the floor after it becomes untenable})$, with 95% CI.

Chart

Table



Floors that fail. For each floor, the share of nights on which someone is still there after it has become untenable, with its interval. The fire floor (14/F) and the floors just above it fail most often.

Where stairs queue in the worst 5%

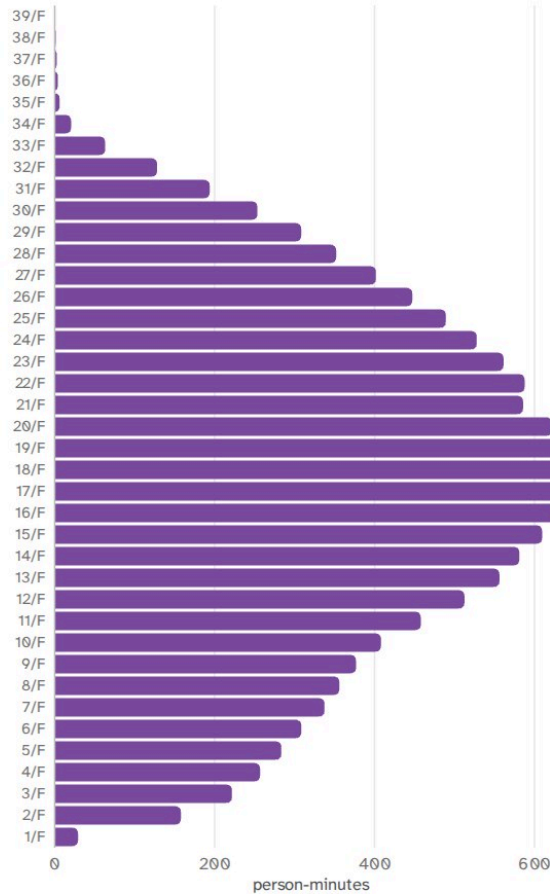
Chart Table

Average person-minutes spent queueing on each flight, by floor (same scale for every stair).

Stair A



Stair B



Where stairs queue in the worst 5%. Person-minutes of queueing on each flight, one chart per staircase, all on the same scale. Here Stair A is lost after 4 minutes, so nearly everyone queues on Stair B, most of all around 15/F to 20/F.

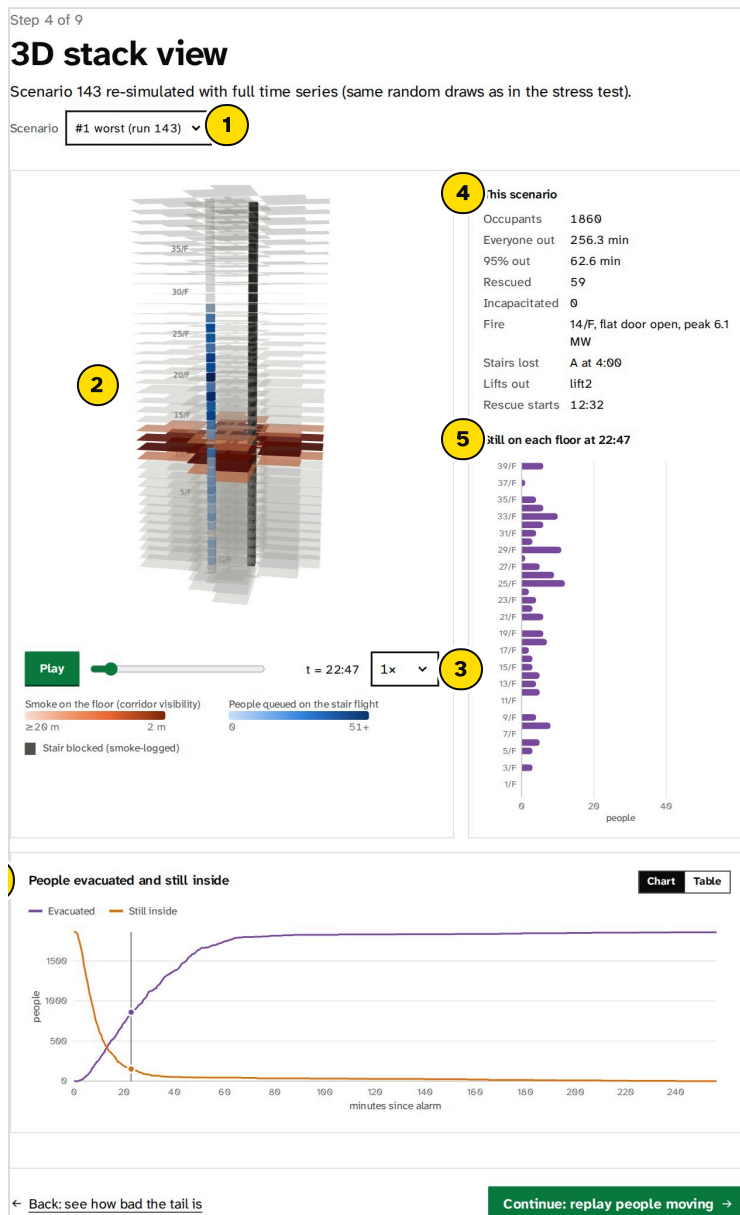
At the very bottom, **Replay a scenario in 3D** lists the ten worst nights and the median night. Click one to watch it in step 4. (In the browser version, the worst night and the median night are recorded.)

HOW TO READ IT

Compare $CVaR_{95}$ with the mean. If the average of the worst 5% is much larger than the mean, a few very bad nights are hiding behind a comfortable average. Then look at who is still inside and where the stairs queue: that tells you whom the bad nights hurt, and where.

Step 4: Watch it floor by floor

The 3D view replays one night, with the same people, fire and broken lift as in the stress test, so you can see how a bad night unfolds: where the smoke goes, where the queues build, and who is left behind.



The worst of the 300 nights, 22 minutes and 47 seconds after the alarm.

What to look for

- Press **Play** and watch the smoke appear on the fire floor and climb to the floors just above it.
- Watch the queue build on the one staircase that is left after Stair A is lost.
- Look at the long, flat end of the orange “still inside” line: those are people waiting for the firefighters. On this night everyone was out only after about four hours.

- 1 Which night** to replay: one of the ten worst, or the median.
- 2 The tower.** Each floor is a thin slab, coloured by the smoke in its corridors: pale is clear, dark brown means you can see only 2 m. The columns are the staircases, coloured by how many people queue on each flight: pale blue is none, dark blue is many. A dark grey column is a smoke-logged staircase. Drag to turn the tower, scroll or pinch to zoom, and point at a floor for its numbers.
- 3 Play / Pause**, the time slider and the playback speed.
- 4 This scenario:** how many people, when everyone was out, when 95% were out, how many were rescued or incapacitated, and what happened that night (the fire, the lost staircase, the broken lift, when rescue started).
- 5 Still on each floor** at the time on the slider.
- 6 People evacuated and still inside** over the whole night. The dots follow the slider.

Step 5: Replay people moving

A closer look at one night: every person is a dot on the floor plan, walking, queueing and taking the stairs.

Step 5 of 9

Replay, person by person

The microscopic engine re-runs scenario 143 with every person as a disc on the floor plan — same people, reaction times and route choices as the fast engine, but physical queues at doors and on stairs.

Scenario **#1 worst (run 143)** **1**

Fast engine vs person-by-person, this scenario
1891 of 1860 people walk; households waiting for a lift or for rescue keep their times from the fast engine.

	Fast (meso)	Person by person	Difference
Half of the walkers out	24.6 min	25.3 min	+0.7 min
95% of the walkers out	58.6 min	58.0 min	-0.6 min
Last walker out	87.3 min	79.8 min	-7.5 min

14/F at 7:00 - fire floor **3** Floor **14/F** **5** People on each floor at 7:00
Click a bar to watch that floor.

4 **5** People on this floor over time. Click a bar to watch that floor.

6 People on this floor over time.

People on 14/F over time **Chart** **Table**

← Back: watch it floor by floor **Continue: find what causes it** →

The fire floor, 7 minutes after the alarm, on the worst night.

Reading the dots

- **Filled purple dot:** a person walking.
- **Hollow purple ring:** a person on the stairs (drawn across the stair enclosure).
- **Grey dot:** a person not moving yet, or waiting (for example for a lift or for rescue).

People who wait for a lift or for the firefighters are not walked by this simulator; they keep the times from the fast simulator. The comparison table counts only the people who walk.

▷ **TRY IT**

Watch the fire floor first: its households react quickly and leave early. Then click the bar for a high floor, such as 35/F, and play again at 15x: you will see people waiting because the staircase below them is full.

Step 6: Find what causes it

Which staircase, door or exit makes the worst nights bad? This screen ranks them by testing each one (chapter 12).

Step 6 of 9

Bottlenecks

Each candidate element (a whole stair, its doors, an exit, the places where queues form) is given more capacity — or a blocked stair is kept usable — and the tail scenarios are re-simulated with the same random draws. The ranking is by how much C_{VaR}₉₅ falls.

Outcome: 95% of occupants | Extra capacity: +50% | **Rank bottlenecks** 1

Losing Stair A drives the tail: keeping it usable cuts C_{VaR}₉₅ of p95 occupant time by 12.4 min (95% CI 11.3 to 13.3 min).
Maximum egress flow 1.62 persons/s · minimum cut: Stair A, 1/F → G/F; Stair B, 1/F → G/F

Change in C_{VaR}₉₅ of 95% of occupants
Minutes, with 95% CI. Click a row to see the element in plan and in 3D.

Chart | Table

Rank	Element	Δ C _{VaR} ₉₅ (min)
1	Keep Stair A usable (no blockage)	-12.4
2	Stair B (all flight)	-8.7
3	Stair A (all flight)	-9.5
4	Stair B doors on e...	-9.5
5	Stair A doors on e...	-9.5
6	Stair B discharge ...	-9.9
7	Main entrance (N) ...	0.0
8	Main entrance (S) ...	0.0
9	Stair A discharge ...	0.0

Δ C_{VaR}₉₅ (min): negative = better

4 Keep Stair A usable (no blockage) Floor: 9/F

Highlighted: heavy outline in the plan; outlined stair column or round markers in 3D.

Where queues form in the worst 5%
Recurrence: share of tail scenarios in which the queue reaches 10 people. Click a row to locate it.

Where	Recurrence	Person-min (tail)	Peak queue
Stair B, 18/F → 17/F	100%	637	46
Stair B, 19/F → 18/F	100%	636	46
Stair B, 17/F → 16/F	100%	635	45
Stair B, 16/F → 15/F	100%	628	45
Stair B, 20/F → 19/F	100%	623	47
Stair B, 15/F → 14/F	100%	612	47
Stair B, 22/F → 21/F	100%	587	47
Stair B, 21/F → 20/F	100%	585	49
Stair B, 14/F → 13/F	100%	582	45
Stair B, 23/F → 22/F	100%	562	47

← Back: replay people moving | Continue: test operational fixes →

The bottlenecks of the 40-storey example, for the time until 95% of people are out.

- Outcome** to improve, **Extra capacity** to give each candidate (+25%, +50% or +100%), and **Rank bottlenecks**.
- The headline**, then the maximum egress flow and the minimum cut.
- The ranking**. Each bar is how many minutes the average of the worst 5% would fall if that element were improved; bars to the left mean an improvement. The whisker is the confidence interval.
- Where it is**. Click a row: the element is outlined on the plan and in the 3D tower. Choose a floor from the menu.
- Where queues form in the worst 5%**: the share of bad nights with a queue of 10 or more people, the person-minutes of queueing, and the longest queue. Click a row to find it.

Do this

- 1 Choose the outcome. **95% of occupants** is a good start, because it is about the crowd rather than the few people waiting for rescue.
- 2 Choose the extra capacity, or keep +50%.
- 3 Click **Rank bottlenecks**. In the full app this re-runs the bad nights for every candidate, which can take a few minutes.
- 4 Read the headline, then click the top rows to see where they are.

What the example shows

“Losing Stair A drives the tail: keeping it usable cuts CVaR95 of p95 occupant time by 12.4 min (95% CI 11.3 to 13.3 min).” More room on Stair B would help next (8.7 minutes). Wider doors or exits would hardly change anything. The building can get out at most 1.62 people per second, limited by the bottom flight of each staircase.

WATCH OUT

A ranking tells you what matters most, not what to build. “Stair B, +50%” means “if Stair B could carry 50% more people”. Keeping a staircase free of smoke, or its doors clear, may achieve much of that without any building work. Discuss it with a fire engineer.

Step 7: Test operational fixes

Let TailSafe search for the best combination of cheap fixes, and check it on fresh nights (chapter 13).

Optimise

Candidate plans are simulated on the same sampled scenarios (common random numbers), combined greedily, then the best plan is confirmed against the baseline on fresh scenarios. Only the confirmation is reported as significant or not.

Minimise	Of	Max wardens	Scenarios per plan	Confirmation scenarios
CVaR ₉₅ (tail) ▼	Everyone out ▼	2	60 ▼	200 ▼

Levers the plan may use

☒ Evacuation lifts
 ☒ Hold stair doors open
 ☒ Stair assignment by floor
 ☒ Phased release
 ☒ Floor wardens

Find a better plan

The settings: what to improve, how many wardens are allowed, how many nights to use, and which fixes the plan may use.

Do this

- 1 Minimise:** choose what to improve: $CVaR_{95}$ (tail), $P(RSET > ASET)$, or $Mean + CVaR_{95}$, and **Of** which outcome.
- 2 Max wardens:** how many floor wardens the plan may use (0 to 10).
- 3 Scenarios per plan:** how many nights each candidate is tested on while searching (30, 60, 100 or 200). **Confirmation scenarios:** how many fresh nights the winner is checked on (100, 200 or 400). More nights are more reliable but slower.
- 4 Levers the plan may use:** untick any fix that is not possible in your building.
- 5** Click **Find a better plan**. In the full app this can take several minutes.

In the browser version, the recorded search used the standard settings: the tail of the time until everyone is out, 60 nights per plan, 200 fresh nights, at most 2 wardens, all five fixes.

Reading the result

Recommended plan

Objective: minimise CVaR₉₅ of total time · 31 plans evaluated

- Use the non-firefighting lifts to evacuate mobility-impaired residents (nearest waiting floor first).
- Station floor wardens on 14/F, 35/F.

Confirmed on 200 fresh scenarios

Paired comparison on identical scenarios; changes with 95% confidence intervals.

CVaR ₉₅ of	Baseline	With plan	Change	95% CI	Verdict
Time until everyone is out	201.5 min	102.0 min	-99.6 min	-107.2 to -91.8	✓ better
Time until the last self-evacuee is out	108.3 min	102.0 min	-6.4 min	-11.8 to -0.7	✓ better
Time for 95% of occupants to get out	66.8 min	75.4 min	+8.7 min	+6.7 to +10.0	△ worse
P(RSET > ASET)	0.58	0.55	-0.040	-0.070 to -0.015	✓ better

⚠ **Trade-off:** this plan makes 95% of occupants significantly worse. Try another objective or restrict the levers before adopting it.

The recommended plan in plain English, and the check on 200 fresh nights. Each row compares the average of the worst 5% without and with the plan, the change, its confidence interval and the verdict.

Here the plan is to use the non-firefighting lifts for mobility-impaired residents (nearest waiting floor first) and to station floor wardens on 14/F and 35/F. On the fresh nights:

- the time until **everyone** is out drops from 201.5 to 102.0 minutes: **better**;
- the time until the **last self-evacuee** is out drops from 108.3 to 102.0 minutes: **better**;
- the time for **95% of occupants** rises from 66.8 to 75.4 minutes: **worse**;
- P(RSET > ASET) drops from 0.58 to 0.55: **better**.

The orange **Trade-off** warning points out the outcome that got worse. Frail residents who would otherwise have walked now wait for the lifts, and on nights when one of the evacuation lifts is broken, the other cannot keep up.

Before / after

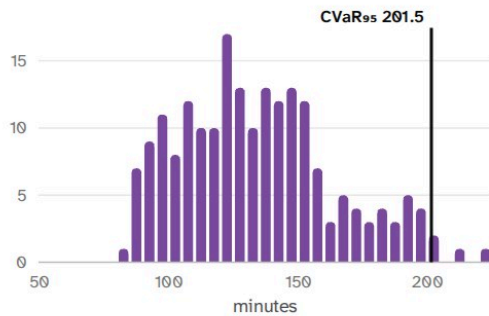
Everyone out

Last self-evacuee

95% of occupants

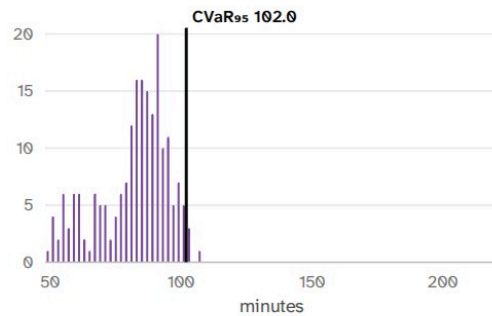
Baseline

Time until everyone is out — 200 identical scenarios



With plan

Time until everyone is out — 200 identical scenarios



The worst confirmation scenario, side by side

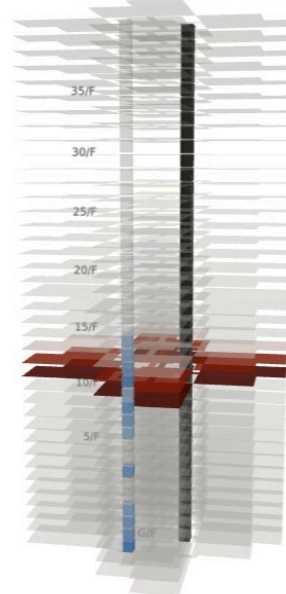
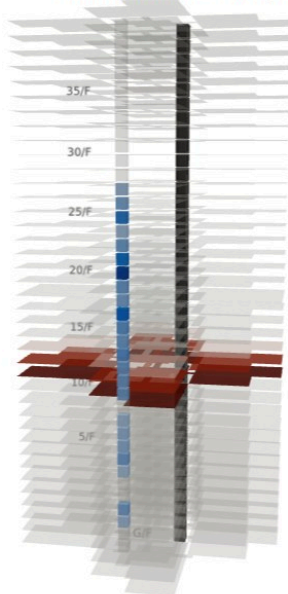
Same fire, same residents, same random draws — only the plan differs.

Baseline

117 still inside · everyone out at 221:13

With plan

167 still inside · everyone out at 60:18



Play

Smoke on the floor (corridor visibility)
≥ 20 m 2 m

People queued on the stair flight
0 58+

Stair blocked (smoke-logged)

t = 25:12

Before / after. Top: the same 200 nights without and with the plan; switch the outcome with the buttons. Bottom: the worst of those nights replayed without and with the plan, side by side on one clock (click **Load animation**, then **Play**). At 25 minutes, more people are still inside *with* the plan, which is the trade-off; yet everyone is out after 60 minutes instead of 221.

Step 8: Try changes instantly

Move the controls and see an estimate straight away, from the quick guesser ([chapter 14](#)). Then check it with the real simulator.

Step 8 of 9

What-if (live)

A graph neural network trained on simulator runs estimates the evacuation-time distribution in milliseconds. Use it to explore; confirm anything you rely on with the full simulation.

Weekday day Weekday night

Weekend day Weekend night

Residents aged 65+ 22%

Fire floor 14/F

☒ Smoke spread and tenability

Staircase smoke-logged Stair A

...from 4 min

Lifts out of service 1

☐ Evacuation lifts for mobility-impaired residents

☐ Wardens on the fire floor and the floor above

Evacuation time, instantly

Estimated in 332 ms. Bar: median to P95; ticks: P75, P90; diamond: CVaR₉₅ (average of the worst 5%).

Chart Table

Surrogate (instant) Simulator (300 runs)

Everyone out 188 218

Last self-evacuee 196 112

95% of occupants 69 66

minutes

3 Confirm with full simulation **4 Open the full results →**

300 Monte Carlo runs with the same settings; the surrogate's estimate stays on the chart for comparison.

5 Where queues are expected

Estimated average person-minutes queueing, most congested first.

Chart Table

Stair	Person-minutes
Stair B, 7/F → 6/F	495
Stair B, 6/F → 5/F	495
Stair B, 3/F → 2/F	495
Stair B, 8/F → 7/F	494
Stair B, 5/F → 4/F	494
Stair B, 9/F → 8/F	492
Stair B, 4/F → 3/F	492
Stair B, 10/F → 9/F	399

person-minutes

6 Trained on 197 slab, 199 cruciform, 209 care home, 195 twin core cases. On a building type it had not seen, its P95 of "everyone out" was off by 33% (care home), 16% (cruciform), 23% (slab), 9% (twin core) on average.

← Back: test operational fixes Continue: get the one-page briefing →

The What-if screen after **Confirm with full simulation**.

The estimate updates a moment after you move a control, and the chart says how many milliseconds it took. In this example the quick guesser puts the average of the worst 5% for "everyone out" at 188 minutes; the full simulation says 218. That gap of about 14% is fine for exploring, but not for deciding.

When the guesser is out of its depth

If your settings go beyond what the network was trained on, an orange box says so and lists why:

- The controls:** time of day, share aged 65+, fire floor, smoke, a smoke-logged staircase and from when, lifts out of service, evacuation lifts, and wardens on the fire floor and the floor above.
- The estimate** for each outcome, in purple. The bar runs from the median to P95; the small ticks are P75 and P90; the diamond is CVaR₉₅. Once confirmed, the simulator's answer appears below it in orange.
- Confirm with full simulation** runs a real stress test of 300 nights with the same settings.
- Open the full results →** shows those results in step 3.
- Where queues are expected**, most congested first.
- How the network was trained**, and how far off it was on building types it had not seen.

 **Outside what the surrogate was trained on — treat the estimate as a rough guide and confirm it**

- More than one lift out of service (training had at most one).

Here, two lifts are out of service; the training only ever had one.

 WATCH OUT

In the browser version the estimate works for any settings, but **Confirm with full simulation** only works for the reference scenario, because only that one was recorded. For anything else, use the full app.

Step 9: Get the one-page briefing

A one-page summary for a building manager, written only from the numbers TailSafe has worked out, and ready to save as a PDF.

Step 9 of 9

Briefing

A one-page summary for the building manager, written only from the numbers computed here. When an LLM is configured on the server it drafts the text, and every number in its draft is checked against the results; a draft with any other number is rejected and the template briefing is shown instead.

Write again **Template only** **Download PDF** **1**

✓ Written from the results; every number checked against them.

Evacuation stress test: Cruciform public housing block, 40 storeys

Scenario: Weekend night, 22% of residents aged 65+, fire on 14/F, Stair A smoke-logged from 4 min, 1 lift out of service. Based on 300 simulated runs of this scenario, with about 1825 synthetic residents in each.

What we found

- On an average run, everyone is out after 136.5 min (including fire-service rescue). In the worst 5% of runs it takes 218.1 min on average (95% confidence 205.7 to 229.4 min).
- The last person who can leave unaided is out after 86.2 min on average, 112.0 min in the worst 5%.
- In 60% of runs someone is still on a floor after its corridors become untenable, or breathes a harmful dose of smoke.
- Floors most often affected: 14/F (58% of runs), 15/F (47% of runs), 16/F (24% of runs).

Who is most at risk

- Households whose most dependent member is a wheelchair user, on 30/F–39/F, are 1.2% of occupants but 100% of the people still inside late in the worst 5% of scenarios.
- Households whose most dependent member is a frail older adult (80+), on 30/F–39/F, are 4.9% of occupants but 76% of the people still inside late in the worst 5% of scenarios.

Why

- Losing Stair A drives the tail: keeping it usable cuts CvA95 of p95 occupant time by 12.4 min (95% CI 11.3 to 13.3 min).
- More capacity at Stair B (all flights) would cut the worst-5% average of the time for 95% of occupants to get out by 8.7 min.
- More capacity at Stair A (all flights) would cut the worst-5% average of the time for 95% of occupants to get out by 0.5 min.

What to do

- Use the non-firefighting lifts to evacuate mobility-impaired residents (nearest waiting floor first).
- Station floor wardens on 14/F, 35/F.

Tested on 200 fresh runs against the same runs without the plan:

- Time until everyone is out, including fire-service rescue (worst 5%): 201.5 min → 102.0 min, better by 99.6 min.
- Time until the last person who can leave unaided is out (worst 5%): 108.3 min → 102.0 min, better by 6.4 min.
- Time for 95% of occupants to get out (worst 5%): 66.8 min → 75.4 min, worse by 8.7 min.
- Runs where someone is caught by smoke: 58% → 55% (better).

Trade-off: the plan makes the time for 95% of occupants to get out worse. Weigh this before adopting it.

Limits

- These results come from a simulation with many assumed parameters; they show where the risk concentrates and what helps, not exact times.
- TailSafe is an educational and decision-support prototype. It is not a substitute for a registered fire engineer, the Fire Services Department, or compliance with the Buildings Department's Code of Practice for Fire Safety in Buildings. All occupants are synthetic; no personal data is used.

← Back: [try changes instantly](#)

The briefing for the 40-storey example, after steps 6 and 7.

Steps 6 and 7 are optional, but the briefing is much more useful with them. The PDF adds a chart of the time until everyone is out, before and after the plan. **Remember:** the briefing inherits every assumption behind the numbers. Share it as a starting point for a conversation with a fire engineer, never as a verdict on a building.

1 Write briefing (then **Write again**) uses the AI writer if the full app has one set up, and the fixed sentences otherwise. **Template only** always uses the fixed sentences. **Download PDF** saves the page.

2 The check: every number in the text was checked against the results. If an AI draft was rejected, an orange note says why.

3 The briefing.

What is in it

- Scenario** and how many nights it is based on.
- What we found:** the average and the worst 5%, how often smoke catches someone, which floors.
- Who is most at risk.**
- Why:** the top bottlenecks (if step 6 was run).
- What to do:** the plan and its tested effect, with any trade-off (if step 7 was run).
- Limits:** the assumptions and the responsible-use notice.

Reading a floor-plan image

No building file? The full app can turn a picture of a typical floor plan into a TailSafe building. You help it by correcting what it gets wrong.

TailSafe

PROTOTYPE Decision support only – not a fire-safety assessment. All residents are synthetic. [Read the notice](#)

1 Set up

Choose the building

Set the scenario

2 Understand the worst nights

See how bad the tail is

Watch it floor by floor

Replay people moving

Find what causes it

3 Make it safer

Test operational fixes

Try changes instantly

Get the one-page briefing

Step 1 of 9

Which building?

Start from one of four Hong Kong building types, read a floor-plan image, or upload a building file (schemas/building.schema.json). Every resident in it will be synthetic.

Building type **Floor-plan image** **Upload file**

Choose File No file chosen [Try a sample plan](#)

PNG, JPEG or PDF (first page). Plans stay on this machine's server.

Select **Reference line** **Add doorway**

Click a room to change its type, or a doorway to delete it.

1. Scale

Reference line is m long

2. Read again

20.0 px per metre (from your line)

3. Check and correct

⚠ No doorway to the outside found; exits will be assumed at the foot of each stair.

⚠ 12 enclosed space(s) without a doorway (shafts, ducts) are ignored.

Room R1 - 45 m² - 1 doorway(s)

Flat / room

4. Build

Storeys (including G/F)

Name

[Build building](#)

The floor-plan editor with the sample plan: rooms coloured by type, doorways marked, and the panel with the four stages on the right.

Do this

- 1** In step 1, click **Floor-plan image**.
- 2** Choose a file: a PNG or JPEG picture, or a PDF (TailSafe uses its first page). Or click **Try a sample plan** to practise.
- 3** **1. Scale.** Click the **Reference line** tool, then click both ends of something whose length you know, such as a 10 m grid line. Type its real length in the box. Without a line, TailSafe guesses the scale from the thickness of the walls, which is less reliable.
- 4** Click **2. Read plan**. TailSafe finds the rooms and doorways and colours the rooms by type.

- 5 **3. Check and correct.** With the **Select** tool, click a room and change its type in the menu: Flat / room, Corridor, Lobby, Staircase, Refuge area or Not walkable. Click a doorway to see its width, and **Delete doorway** if it is wrong. To add a missing doorway, choose **Add doorway** and click one side of the opening, then the other. Warnings in orange tell you what to check.
- 6 **4. Build.** Type the number of **storeys (including G/F)** and a name, then click **Build building** →. The same plan is used for every floor, and doorways to the outside become exits on the ground floor.
- 7 Check the new building on the usual step 1 screen, then click **Confirm building** →.

What the reader can and cannot do

- Walls must run straight across and straight up and down the picture. Turn scanned plans upright first; curved or slanted walls are not read.
- Walls must be drawn thicker than other lines. Plans drawn with thin or hatched walls need more correcting.
- Doorways are gaps between 0.6 and 2.0 m wide. Wider openings join the rooms on both sides into one.
- Refuge floors and lifts are not found; the ground floor usually differs from the typical floor and should be checked.
- If the plan shows no doors to the outside, TailSafe puts an exit at the foot of every staircase, and says so.
- On TailSafe's own test plans it finds 96 to 100% of the doorways. Those plans are drawn by the computer, so real drawings will do worse: checking is part of the job.

The plan is only read on the computer running the full app; it is not sent anywhere else.

Putting it together

A complete example from question to answer, the command line for people who like typing, the limits to keep in mind, and help for when something goes wrong.

30	A worked example: one night at 3 a.m.	59
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33	Troubleshooting	64
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A worked example: one night at 3 a.m.

This chapter follows one question all the way through TailSafe, using a larger study of 1,000 nights run in the full app.

THE QUESTION

A 40-storey public housing block. It is Sunday, 3 a.m. A fire starts in a flat on 14/F. Four minutes later, Stair A fills with smoke. One lift is out of service. 22% of the residents are 65 or older. How bad could the evacuation get, why, and what would help?

This is TailSafe's **reference scenario**: the standard example that the browser version records for every building type, and that the command line uses unless you give it another one. The numbers below come from 1,000 nights with seed 0. The browser version's 300 nights are the first 300 of these, so its numbers are close but not identical.

1. How bad is it?

219 min

Everyone out, average of the worst 5% (212 to 226)

138 min

Everyone out, average of all nights

114 min

Last person out on their own, worst 5%

0.60

P(RSET > ASET): someone caught by smoke on 60 nights in 100

- On an average night, 95% of the residents are out after about an hour (60 minutes); on the worst 5% of nights, after 66.5 minutes. The last person able to leave unaided is out after 86 minutes on average.
- Everyone is out, counting the people carried down by firefighters, after 138 minutes on average and 219 minutes on the worst 5% of nights. The gap between the two shows that a few bad nights hide behind the average.
- Smoke catches someone on 60% of nights, most often on 14/F (58% of nights), 15/F (47%) and 16/F (19%): the fire floor and the floors just above it. On 6.5% of nights at least one person is incapacitated.

2. Who is left?

- Households with a wheelchair user on 30/F to 39/F are 1.2% of the residents but all of the people still inside at the very end of the worst nights. They wait for the firefighters, who come down the stairs one household at a time.
- Among people leaving on their own, households with a frail resident aged 80 or over on 30/F to 39/F are 4.9% of the residents but 70% of the late stragglers.

3. Why?

Losing Stair A. From four minutes on, the whole block must use Stair B, which lets about 48 people a minute down. Queues form on Stair B on every bad night. The bottleneck ranking confirms it: keeping Stair A usable would cut the worst-5% average of the time for 95% of residents to get out by 12.8 minutes (12.2 to 13.4), and giving Stair B 50% more capacity by 8.6 minutes. Wider doors would hardly matter.

4. What helps?

The plan search proposes using the non-firefighting lifts for mobility-impaired residents (nearest waiting floor first), and floor wardens on 14/F and 35/F. Checked on 400 fresh nights:

Average of the worst 5%	Without the plan	With the plan	Change	Verdict
Time until everyone is out	204.9 min	104.6 min	−100.3 min	better
Time until the last self-evacuee is out	109.5 min	104.6 min	−5.0 min	better
Time for 95% of occupants to get out	66.3 min	75.6 min	+9.3 min	worse
Nights on which smoke catches someone	58%	55%	−3 points	better

5. What should a building manager take away?

- 1 The longest nights are about rescue.** Residents who cannot use the stairs wait a long time. Lifts for them halve the tail of the time until everyone is out.
- 2 But there is a price.** With lifts, frail residents who could have walked wait for a lift instead, so the crowd as a whole gets out later. When one of the evacuation lifts is the broken one, the other cannot keep up. This needs weighing, perhaps by limiting the lifts to wheelchair users (a setting TailSafe can test).
- 3 Stair A matters most.** Anything that keeps it usable for longer is worth discussing with a fire engineer.
- 4 All of this rests on assumed numbers** ([chapter 32](#)). It shows where to look and what to ask, not exact minutes.

► TRY IT

Follow the same path in the browser version, with 300 nights: step 3 for the spread, step 4 to watch the worst night, step 6 for the cause, step 7 for the plan, step 9 for the briefing. The numbers differ slightly from this chapter, because they come from 300 nights instead of 1,000.

The command line

Everything in the web app can also be done by typing commands. This is handy for running many studies, or for keeping a record of exactly what was run.

The `tailsafe` command comes with the full app when it is installed from the source code ([chapter 18](#)). After `make install`, either type `source .venv/bin/activate` once in your terminal, or put `.venv/bin/` in front of every command. `tailsafe --help` lists every command, and `--help` after any command lists its options.

Command	What it does
<code>tailsafe building generate cruciform --storeys 40 --out out/tower.json</code>	Makes a building and saves it as a building file.
<code>tailsafe sim run cruciform --slot weekend_night --share-65 0.22 --block-stair A@240 --plot out/run.png</code>	Simulates one single night and draws it. <code>A@240</code> means Stair A is lost after 240 seconds.
<code>tailsafe stress run cruciform --runs 1000 --out out/stress</code>	The stress test: 1,000 nights of the reference scenario. Prints the key numbers and saves the results and a chart.
<code>tailsafe stress report out/stress --loss self_evacuation_time</code>	Prints a saved stress test again, for another outcome.
<code>tailsafe stress bottlenecks out/stress</code>	Ranks the bottlenecks of a saved stress test.
<code>tailsafe optimize cruciform --out out/plan</code>	Searches for a plan and checks it on fresh nights.
<code>tailsafe brief --stress out/stress --optimization out/plan --pdf out/brief.pdf</code>	Writes the one-page briefing (text and PDF).
<code>tailsafe micro run cruciform --index 3 --level 14 --plot out/floor.png</code>	Replays night 3 person by person and draws 14/F.
<code>tailsafe vision detect plan.png --scale 0,0,200,0,10 --out det.json</code>	Reads a floor plan. The scale says that the line from pixel (0, 0) to pixel (200, 0) is 10 metres long.
<code>tailsafe vision build det.json --storeys 30 --out out/plan_building.json</code>	Stacks the read floor into a 30-storey building file.
<code>tailsafe validate</code>	Runs the checks of the simulator against hand calculations.
<code>tailsafe params list --assumptions</code>	Lists every number that still needs a published source.

Your own scenario in a file

The stress test, the plan search and the replays take `--spec` followed by a scenario file, written in YAML (a simple text format) or JSON. For example, save this as `my-night.yaml`:

```

name: weekday-afternoon
time_slot: weekday_day      # or weekday_night, weekend_day, weekend_night
share_65_plus: 0.30         # 30% of residents aged 65 or over
fire_level: 10              # fire on 10/F
stair_blockages:
  - stair: B                # Stair B lost between 2 and 10 minutes in
    time: {dist: uniform, min: 120, max: 600}
lifts_out_of_service: 1
hazard: {enabled: true}     # simulate the smoke

```

and run it with `tailsafe stress run cruciform --spec my-night.yaml --runs 300 --out out/weekday`. Times are in seconds. Anything you leave out takes its standard value.

CHAPTER 32

Limits and responsible use

Every model simplifies the world. These are the most important simplifications in TailSafe, and they are why its results are for learning and comparing, not for exact answers.

- **Many numbers are educated guesses.** 146 of the 169 numbers in the settings file are marked as assumptions, including how fast frail people walk, how long people take to react at night, and how quickly firefighters rescue people.
- **The buildings are generic.** The four building types are made by rules, not measured from real blocks. Real stair widths, door widths and refuge floors may differ.
- **The smoke model is simple:** one fire; each room is one well-mixed box of air; no wind, no sprinklers, no fans that keep smoke out of the stairs. It is not a detailed fire simulation.
- **People behave simply.** Everyone knows the quickest way out, and only learns a staircase is blocked when they reach it. On the stairs, fast walkers can pass slow ones freely, which is optimistic on narrow stairs.
- **Rescue is a rough model.** It is an assumption, and it decides how long “everyone out” takes. Always read it alongside the other two outcomes.
- **Not yet compared with real drills.** TailSafe's checks show that the simulator does what the engineering method says. They do not show that the method and the numbers are right for Hong Kong towers.
- **The quick guesser learns the simulator, not reality,** and is weakest for building types unlike those it learned from, especially care homes.
- **The browser version uses 300 nights,** so its confidence intervals are wider than those of a 1,000-night study.

The full list is in `docs/assumptions.md`, and the checks are described in `docs/validation.md`.

How to talk about the results

Do not say

- “Our building is safe.”
- “Evacuation takes 219 minutes.”
- “TailSafe proved that lifts are the answer.”

Say instead

- “In TailSafe’s model, the worst nights come from losing Stair A.”
- “The model suggests lifts help wheelchair users a lot, with a cost for the crowd.”
- “We should ask a fire engineer about keeping Stair A usable.”

THE RESPONSIBLE-USE NOTICE

TailSafe is an educational and decision-support prototype. It is not a substitute for a registered fire engineer, the Fire Services Department, or compliance with the Buildings Department's Code of Practice for Fire Safety in Buildings. All occupants are synthetic; no personal data is used.

Troubleshooting

Something not working? Find the problem below.

In the web app

A message says the browser version has no results for these settings.

The browser version only holds the reference scenario of each building. Press

Back to the reference scenario, use the What-if screen for an estimate, or run the full app ([chapter 17](#)).

A step in the list is grey and cannot be clicked.

Do the earlier steps first. Most steps need a building (step 1), and the results need a stress test (step 2).

The 3D view is blank.

Your browser may have WebGL, its graphics feature, switched off. Update the browser, try another one, or turn on “hardware acceleration” in its settings.

The charts are cramped on my phone.

Turn the phone sideways, or switch the chart to **Table**.

The What-if screen shows an orange box “Outside what the surrogate was trained on”.

Your settings go beyond what the quick guesser learned. Treat its estimate as rough and press

Confirm with full simulation (in the full app).

The briefing shows a note that a draft was rejected.

The optional AI writer used a number that is not in the results, so TailSafe showed the fixed-sentence briefing instead. Nothing needs fixing.

The floor-plan reader misses rooms or joins them.

Draw a reference line for the scale, check that the walls are thick and straight, then fix room types by hand and add missing doorways ([chapter 29](#)).

With the full app

“docker: command not found”, or “Cannot connect to the Docker daemon”.

Docker is not installed, or not running. Open Docker Desktop and wait until it says it is running, then try again.

“port is already allocated”.

Something else, perhaps another copy of TailSafe, is already using port 7860. Stop it, or start TailSafe on another port with `docker run -p 8080:7860 tailsafe` and open `http://localhost:8080`.

The build stops with a download or certificate error.

Some networks, often at work, block or inspect downloads. Try another network, or ask your IT team.

My phone cannot open the full app.

Use the computer's IP address, not `localhost` (which on the phone means the phone itself). Check that both are on the same Wi-Fi, and allow Docker through the computer's firewall.

The first run is slow.

That is normal: the simulator prepares itself the first time (about ten seconds).

A run takes very long.

Use fewer nights (300 instead of 2,000). Ranking bottlenecks and searching for a plan are the slowest steps; fewer *scenarios per plan* makes the search quicker.

My numbers differ from someone else's.

Compare the seed, the number of nights, the building options and every scenario setting. The same settings with the same seed always give the same numbers.

CHAPTER 34

Questions people ask

Does TailSafe use real residents' data?

No. Every resident is made up by the computer, following general rules about ages, households and behaviour. No personal data is used.

Can TailSafe tell me whether my building is safe?

No. It is a learning and decision-support tool. Only a registered fire engineer, the Fire Services Department and compliance with the Code of Practice can do that.

Is anything I enter sent somewhere?

In the browser version, nothing you enter is uploaded. In the full app, buildings and floor plans go only to the computer that runs it; on a public Hugging Face Space, that is Hugging Face's server.

Why does “everyone out” take hours?

Because a few residents cannot use the stairs and wait for the firefighters, who, in TailSafe's model, rescue one household at a time. How rescue works is an assumption. The other two outcomes show how the stairs themselves perform.

Why are the lifts not used?

In Hong Kong, lifts are normally kept for the firefighters during a fire, so TailSafe's standard scenario does not use them. Using some lifts to evacuate residents who cannot manage the stairs is one of the fixes you can test.

Why did my results change when I changed the seed?

A different seed means different dice, so different nights. With enough nights the change is small, and the confidence intervals tell you how small.

How many nights should I run?

300 to explore; 1,000 or more for comparisons that matter.

Why is the What-if estimate different from the full simulation?

It is a quick learned guess. On buildings like those it trained on it is usually within 10 to 20%; on unfamiliar buildings it can be further off. That is why the confirm button exists.

Can I use TailSafe for an office, a school or a shopping centre?

Not as it is. TailSafe is built for homes: households, who is at home at different times of day, and residents' ages. Other buildings would need a different model of the people in them.

Where do TailSafe's numbers come from?

From one settings file, `config/params.yaml`. Every number there says where it comes from, or that it is still an assumption. `tailsafe params list --assumptions` lists the assumptions.

Where do I report a problem or suggest an idea?

On the project's GitHub page, github.com/chingkheinganba231005/Tail_Safe, under *Issues*.

Word list

Able adult

An adult who can walk and help others.

Alarm

The signal that starts the evacuation. In TailSafe everyone hears it when the fire starts.

ASET

Available safe egress time: how long a place stays safe before smoke makes it untenable. The smoke clock.

Average

See *mean*.

Bottleneck

A narrow place, such as a staircase or door, that slows everyone behind it.

Browser version

The TailSafe website anyone can open, with results recorded in advance for four standard buildings.

Capacity

How many people can pass through a door or down a staircase each second.

CI (confidence interval)

A range that we can be fairly (95%) sure contains the true value. More nights give a narrower range.

Common random numbers

Testing different plans on exactly the same nights, so that only the plan differs.

Container

A sealed box that holds a program and everything it needs. Docker runs containers.

Counter-flow

Walking the opposite way to everyone else, for example back upstairs to fetch a relative.

Counterfactual

A “what if it had been different?” test, such as re-running a night with a wider staircase.

CVaR₉₅

The average of the worst 5% of nights. TailSafe's main measure of how bad the bad nights are.

Docker

A free program that runs software in containers. The easiest way to run the full app.

Domestic helper

A live-in helper who can help an older person or carry a wheelchair user.

Egress

Going out; leaving a building.

Evacuation

Everyone leaving a building because of danger.

Evacuation lift

An ordinary lift used during a fire to carry residents who cannot use the stairs. One of the fixes TailSafe can test.

FED

Fractional effective dose: a count of how much smoke and heat a person has taken in. At 1 they are incapacitated; TailSafe treats 0.3 as harmful.

Firefighting lift

The lift kept for firefighters. TailSafe never uses it to evacuate residents.

Floor warden

A trained resident or staff member who knocks on doors and helps one household that cannot use the stairs.

Full app

TailSafe running on your own computer: the simulator, the server and the web app together.

G/F, 1/F, 14/F

Hong Kong floor names: ground floor, first floor, fourteenth floor.

Graph

A map of places (nodes) joined by connections (edges).

Histogram

A bar chart that counts how many nights fell into each band of times.

Hydraulic model

A standard engineering method that treats crowds a bit like water in pipes, to work out how many people can pass a door or a staircase.

Incapacitated

No longer able to move by oneself.

IP address

A computer's address on a network, such as 192.168.1.23.

Localhost

"This computer", in a web address.

Mean

The average: add up all the values and divide by how many there are.

Median

The middle value: half are smaller, half are bigger.

Mesoscopic

"In the middle": TailSafe's fast simulator, which moves groups along the map with queues at every door and staircase.

Microscopic

Person by person: the slower simulator that walks each person across the floor plan.

Monte Carlo method

Working something out by repeating a random experiment many times and counting the results.

Neural network

A program that learns patterns from examples. TailSafe's learned from 800 simulated buildings.

Night

In this guide, one simulated version of the fire; on screen also called a scenario or a run.

Operational fix

A change to how a building is run, not to its walls.

P95

The 95th percentile: 95 nights in 100 are quicker than this. The border of the worst 5%.

P(RSET > ASET)

The share of nights on which someone is still on a floor after it has become untenable, or breathes a harmful dose of smoke.

Person-minutes

People multiplied by minutes: 10 people queueing for 3 minutes is 30 person-minutes.

Phased release

Letting some floors leave first while others wait.

Port

A numbered door on a computer where a program listens, such as 7860.

Pre-movement time

The time between the alarm and starting to move.

Reference scenario

TailSafe's standard example: Sunday, 3 a.m., fire on 14/F, Stair A smoke-logged after 4 minutes, one lift out of service, 22% of residents aged 65 or over.

Refuge floor

An open floor in a tall building where people can step out of the staircase and rest in fresher air.

Risk ratio

How much more common a group is among the late stragglers than in the whole building.

RSET

Required safe egress time: how long the people in a place need to get out. The walking clock.

Scenario

A description of the fire night to test: the time, the people, the fire, the smoke, the stairs and the lifts.

Scissor staircase

Two staircases twisted around each other in one shaft.

Seed

The starting number for the computer's dice. The same seed gives the same nights.

Smoke-logged

So full of smoke that nobody can use it.

Stack effect

Hot, smoky air rising up staircases and lift shafts, as up a chimney.

Straggler

A household still inside near the end of a night.

Stress test

Running hundreds or thousands of nights and looking at the spread of results.

Surrogate

A fast stand-in for the simulator: TailSafe's quick guesser.

Synthetic

Made up by the computer, not taken from real people.

Tail

The rare, very slow nights at the far end of the spread of results.

Tenable / untenable

Safe to be in / too dangerous to be in.

Winner's curse

The plan that looks best on a set of nights is partly lucky, so it must be checked on fresh nights.

CHAPTER 36

Where to learn more

TailSafe

- **The browser version:** https://chingkheinganba231005.github.io/Tail_Safe/
- **The source code and the project page:** https://github.com/chingkheinganba231005/Tail_Safe
- **In the docs folder:** `architecture.md` (how each part works, in technical detail), `validation.md` (what has been checked, and how well it holds), `assumptions.md` (every simplification, in plain language) and `spec.md` (what TailSafe is meant to do).
- **DEVELOPMENT.md:** commands, conventions and how to publish the two versions.
- **config/params.yaml:** every number TailSafe uses, with its source.

Sources named in TailSafe

TailSafe's settings and documents point to these sources, among others. The settings file marks its published sources as still to be checked against the edition used.

- *SFPE Handbook of Fire Protection Engineering*: the hydraulic model of people's movement, and standard fire growth rates.
- ISO 13571: guidelines for estimating the time available for escape (tenability limits and FED).
- Frantzich and Nilsson (2003): walking speed in smoke.
- Tordeux, Chraïbi and Seyfried (2016): the collision-free speed model behind the person-by-person replay.
- Buildings Department, *Code of Practice for Fire Safety in Buildings* (2011): named as the reference to check for Hong Kong's rules on means of escape.



TailSafe

Evacuation stress-testing for high-rise buildings: how bad the worst nights are, why, and what fixes them.

github.com/chingkheinganba231005/Tail_Safe

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